

universität freiburg

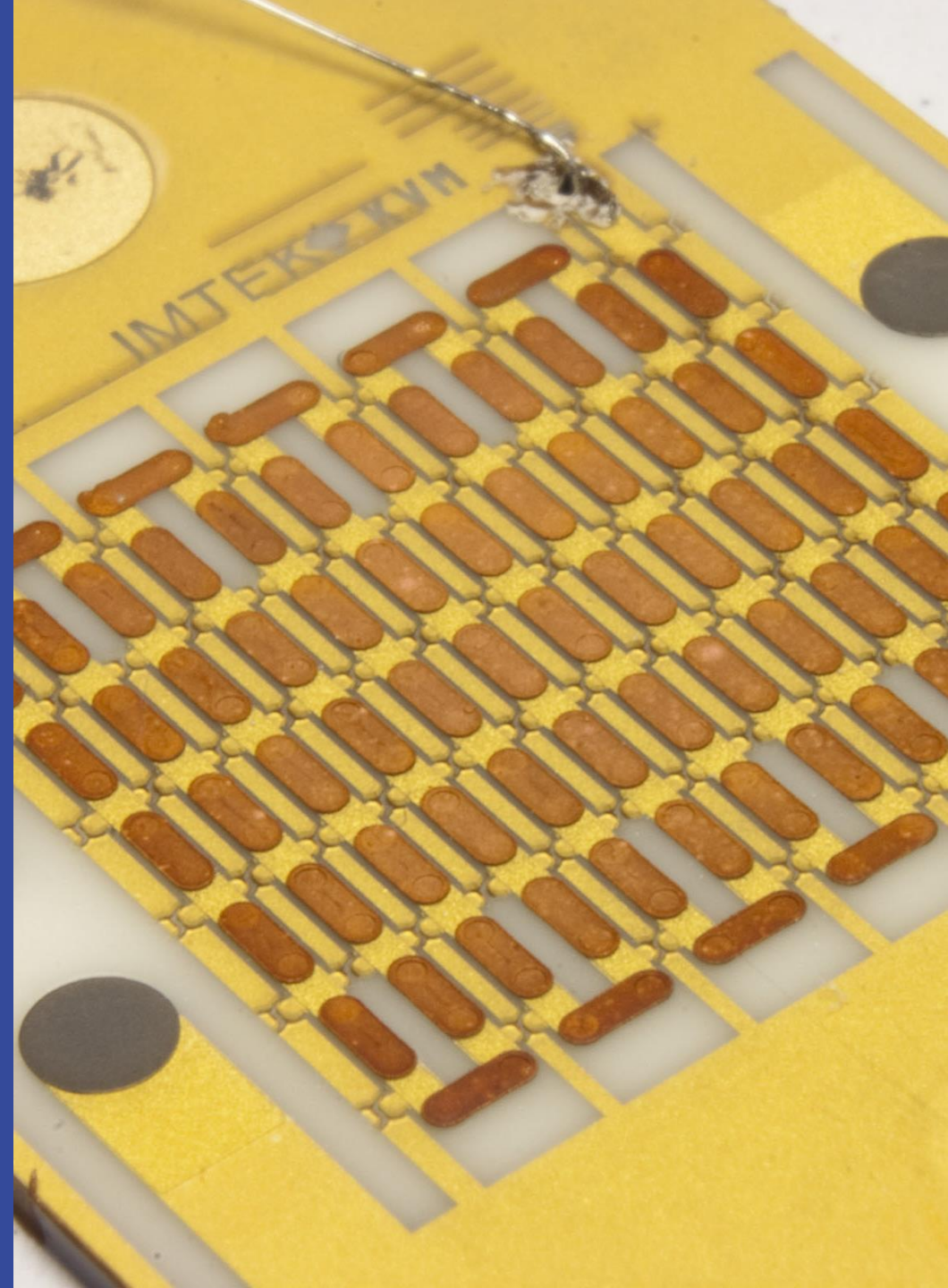
Micro Energy Harvesting

Chapter 1: Introduction

Micro Energy Harvesting SoSe 2026

IMTEK – Lehrstuhl für Konstruktion von Mikrosystemen

Prof. Dr. Peter Woias, Dr. Suman Kundu, Dr. Frank Goldschmidtboing

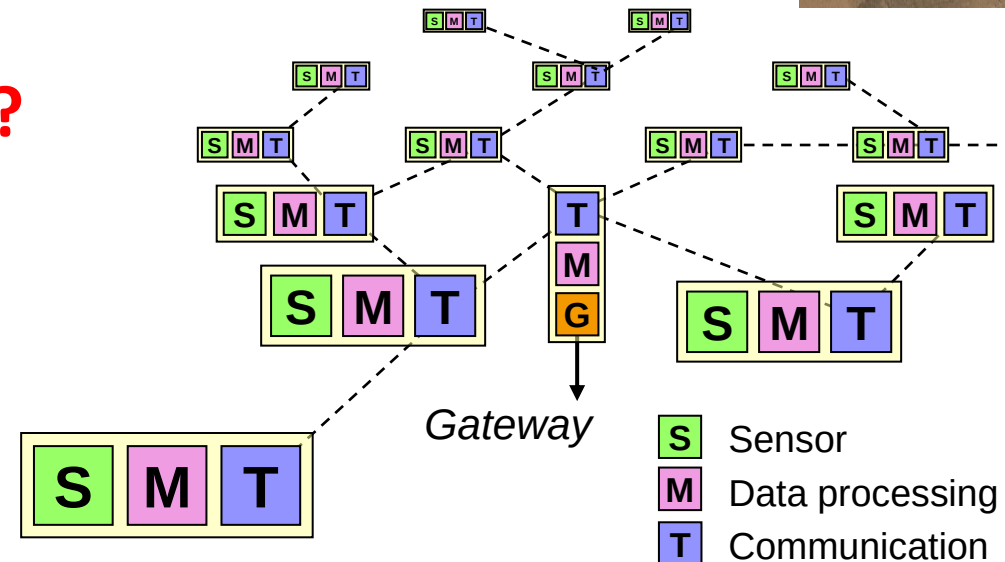


Micro Energy Harvesting

Why?

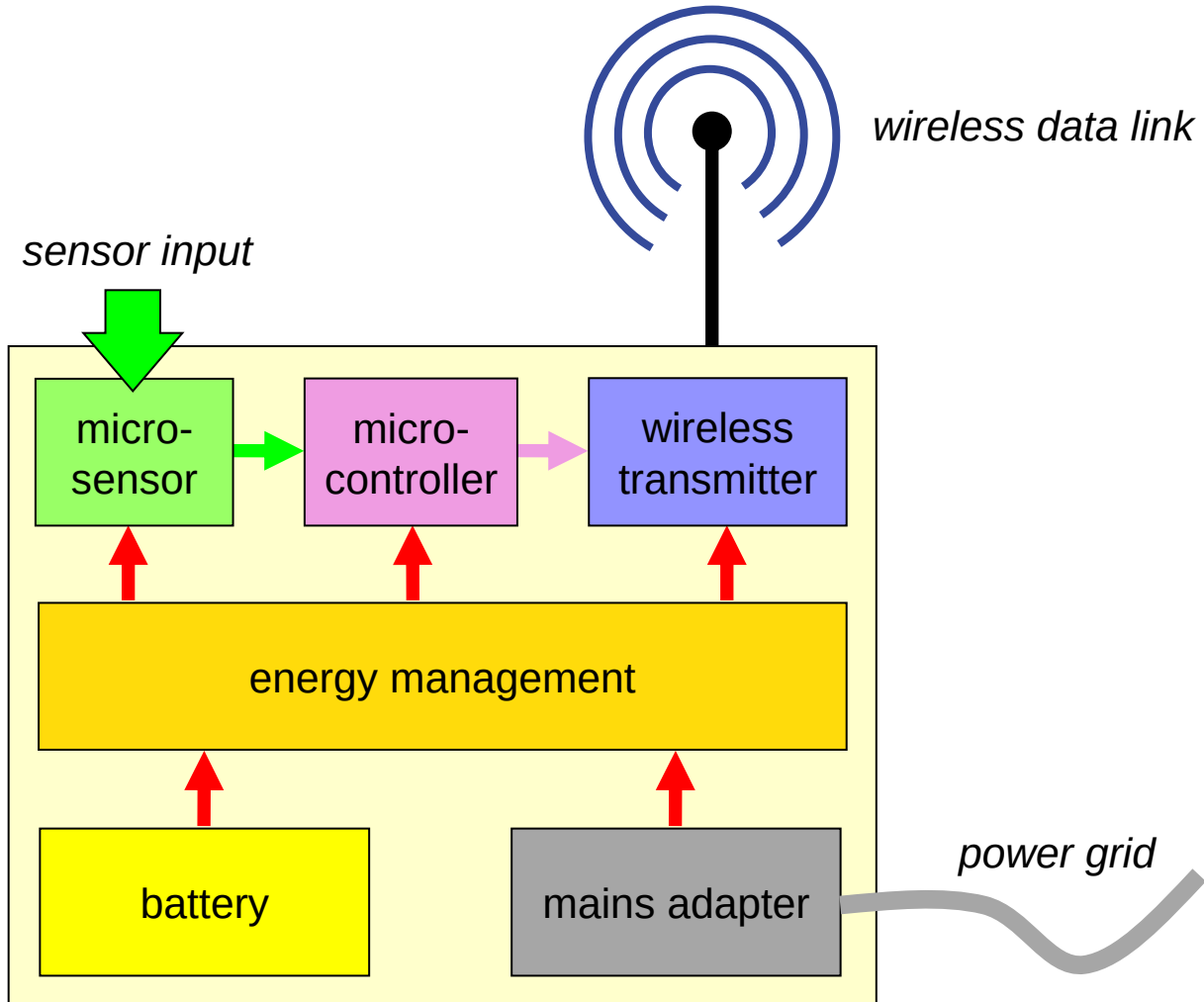


... power supply ?



Micro Energy Harvesting

Power supply of distributed embedded systems...



Battery-based Power Supply



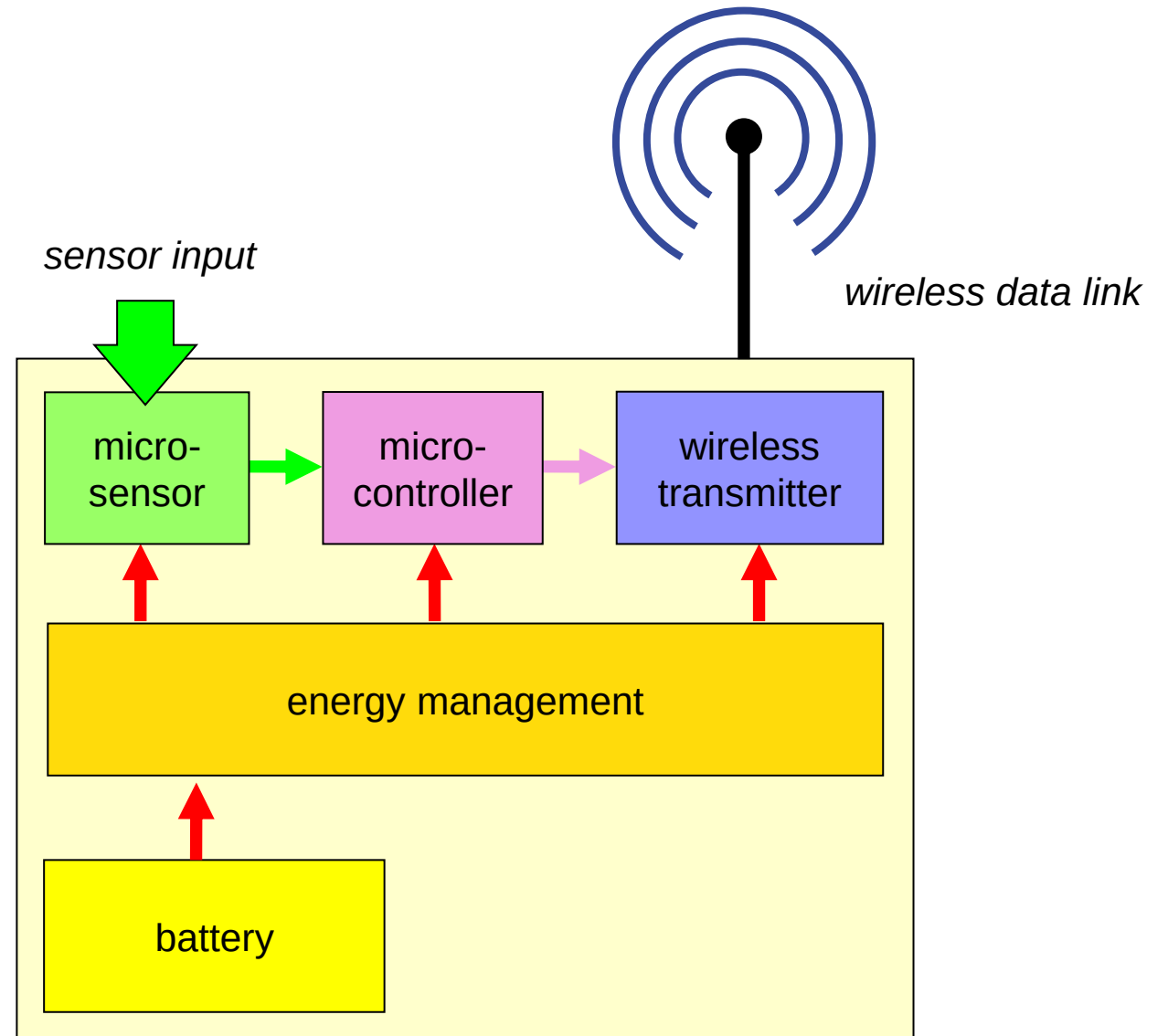
Battery-powered devices
some years ago...



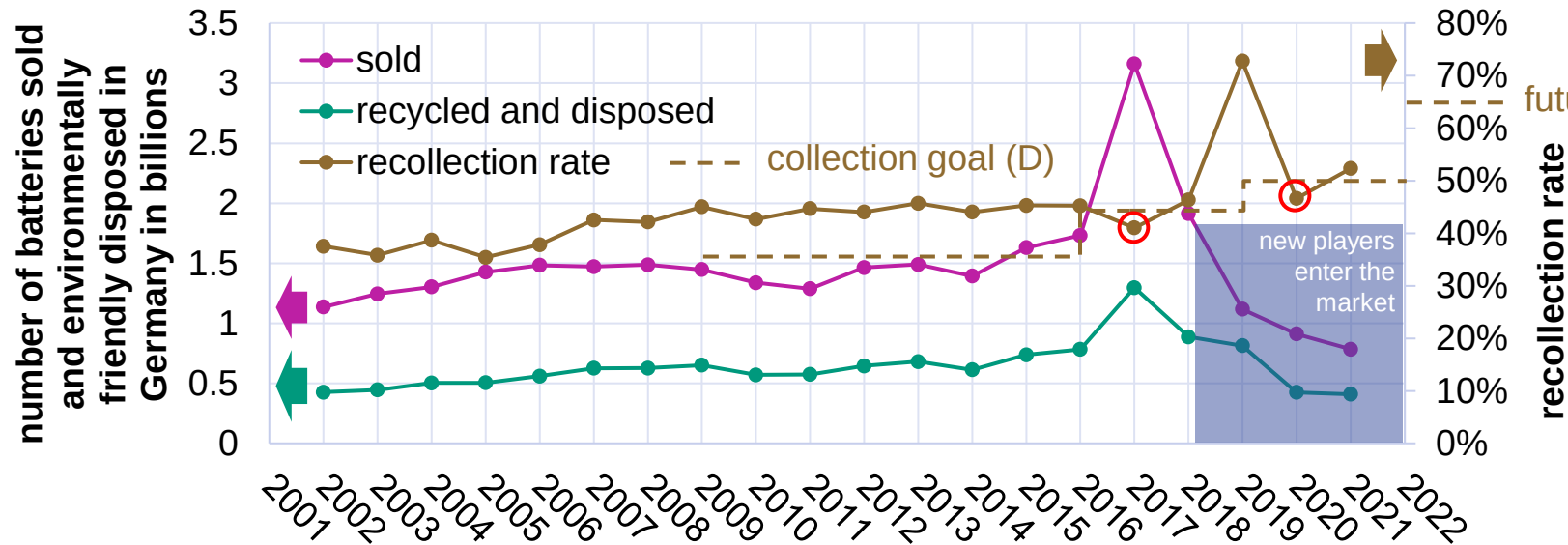
Battery-powered devices
today...

Issues

- limited operation time
- limited operation range
- maintenance costs
- environmental concerns



Battery Consumption and Disposal in Germany



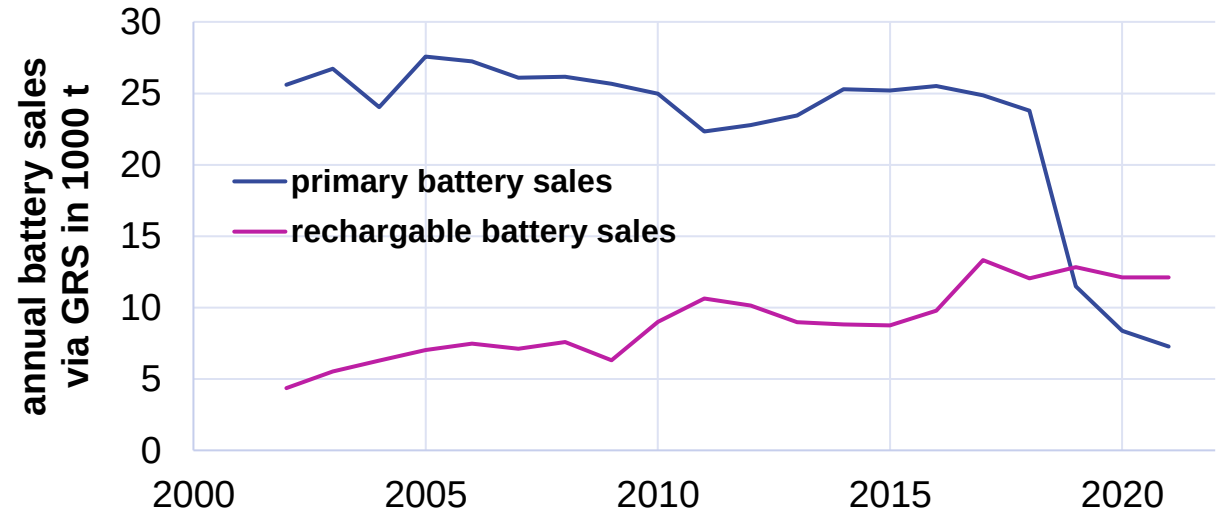
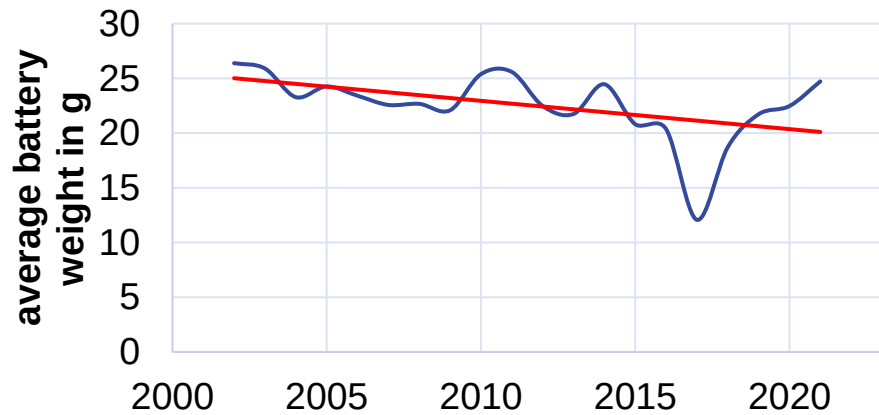
Ref: German battery recycling system „Stiftung Gemeinsames Ruecknahmesystem Batterien“ - GRS

Example: numbers of 2021

- sales through GRS: 19 403 tons = 784 millions of batteries
- disposal through GRS: 10 153 tons = 410 millions of batteries
- „missing batteries“: 9 250 tons = 374 millions of batteries
- average recollection rate: 49.2 % (between 2012 and 2021)



Battery Consumption and Disposal in Germany



All data taken from GRS

Trends

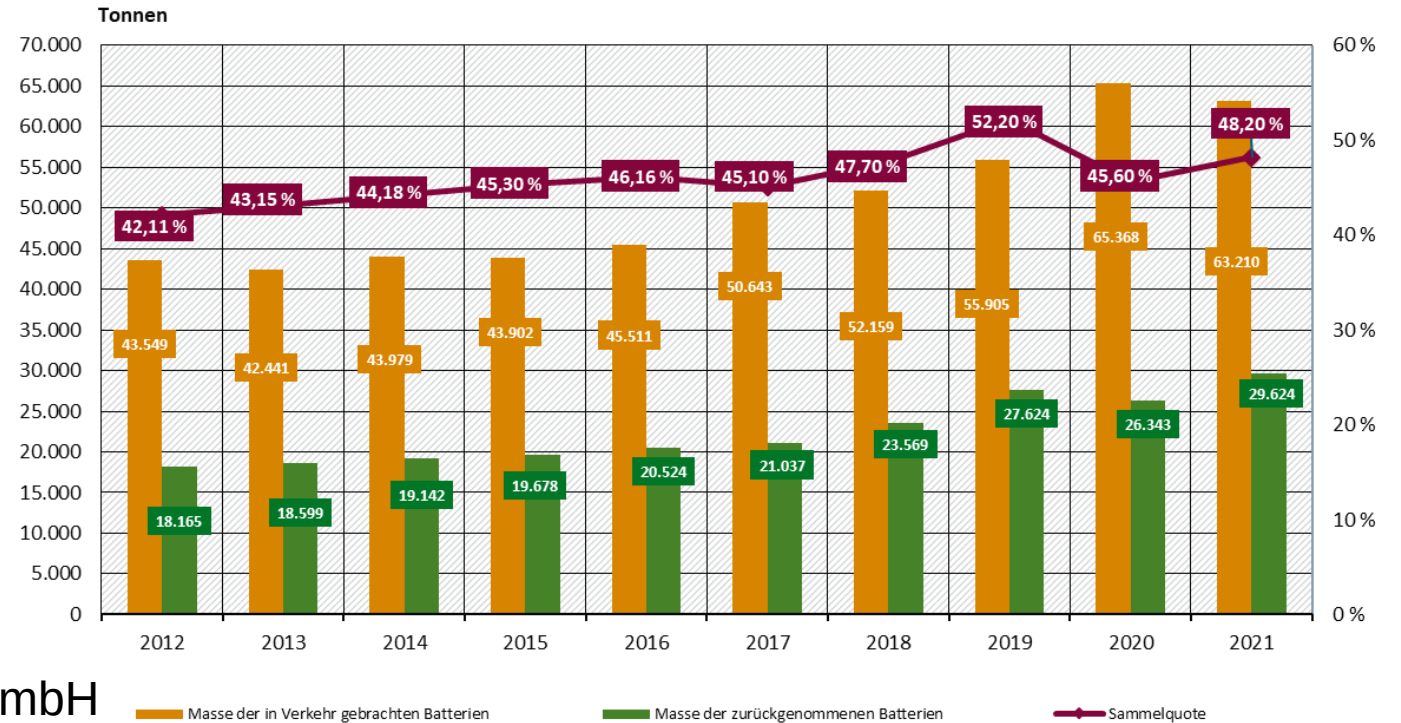
- decrease of average battery weight (caused by smaller consumer batteries)
- increase of environmentally-friendly disposal (although not sufficient)
- significant increase of battery consumption expected (e.g. car batteries)
- Increase in rechargeable batterie sales even though overall sales through GRS dropped

Why do we not see more rechargeable batteries ?

- many rechargeable devices not included (phones, Bluetooth speakers and headphones, ...)

Battery Consumption and Disposal in Germany

- Systems for collecting „used device batteries“ in Germany report annually
- Various players:
 - GRS Batterien (plus sub-systems)
 - CCR REBAT
 - Landbell GmbH
 - DS Entsorgungs- und Dienstleistungs- GmbH
 - Öcorecell
 - ECOBAT Logistics GmbH
- Additionally: used batteries from cars and industry



as much as 33586 t of batteries missing in 2021 alone...

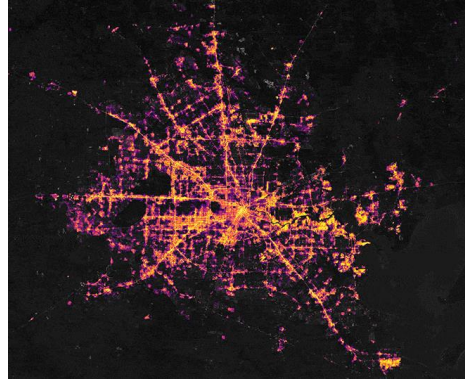
Graphic from:



Grid-based Power Supply



German main power grid
2005/2006

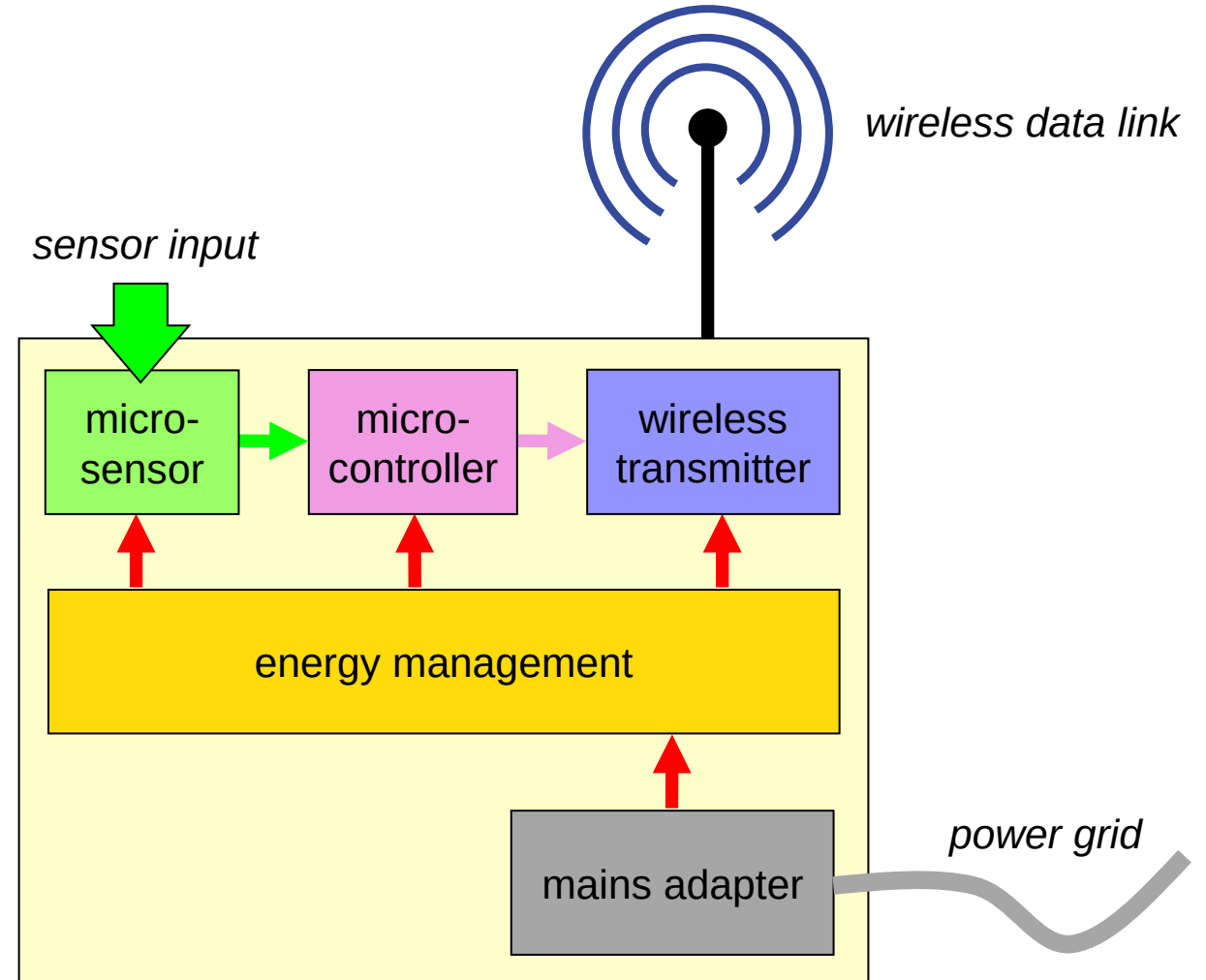


Huston February 2021
Texas power crisis

Issues

„Is there a grid?“ ... if not :

- grid establishment
- grid enlargement
- grid maintenance
- grid failure



On-board power grids



Wiring trips up A380 schedule

Published: WEDNESDAY, JUNE 14, 2006

INTERNATIONAL
Herald Tribune

Airbus A 380: **530 km**



Eurocopter EC 135: **22 km**

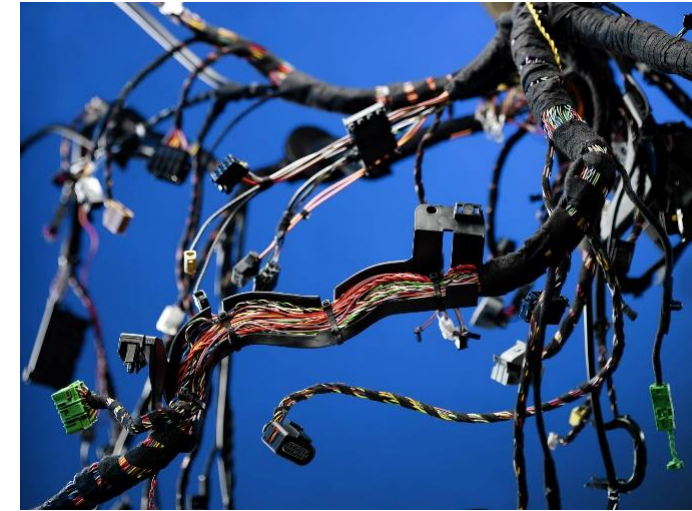
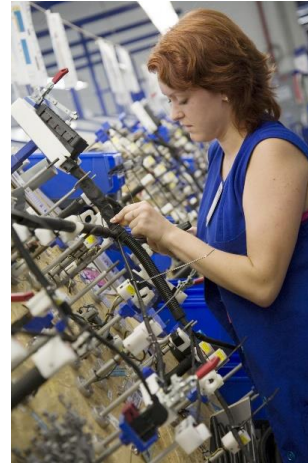


Porsche 911: **3 km**



On-board power grid

Cars



all pictures: © Leoni AG

- large, complex and heavy
mid-class car: ~ 3.000 m of wires make appr. 6% of total car weight
- manual fabrication, time-consuming installation
Power grid virtually „grows“ into the car
 - bad access
 - complex and expensive maintenance



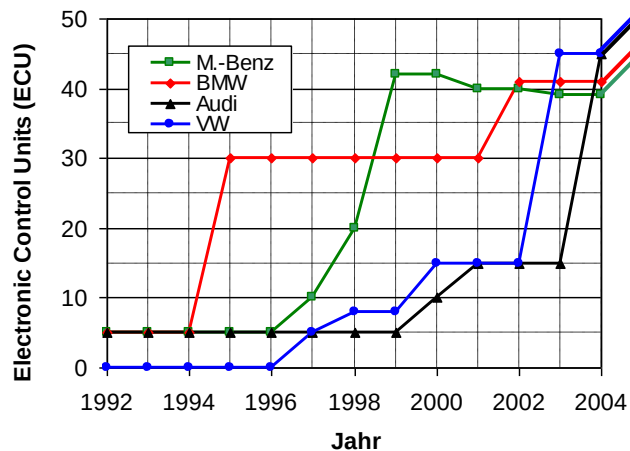
On-board power grid Cars

Evolution of on-board power grids in cars

J. Leohold, Communication Requirements for Automotive Systems, 5th IEEE Workshop on Factory Communication Systems, Vienna, Sept. 22-24, 2004.



VW Passat 3 (1993)



VW Passat 6 (2005)

- up to 150 ECUs in cars in 2019 especially for electric cars
- consolidation efforts to reduce ECUs by e.g. one central unit (cockpit domain controller)

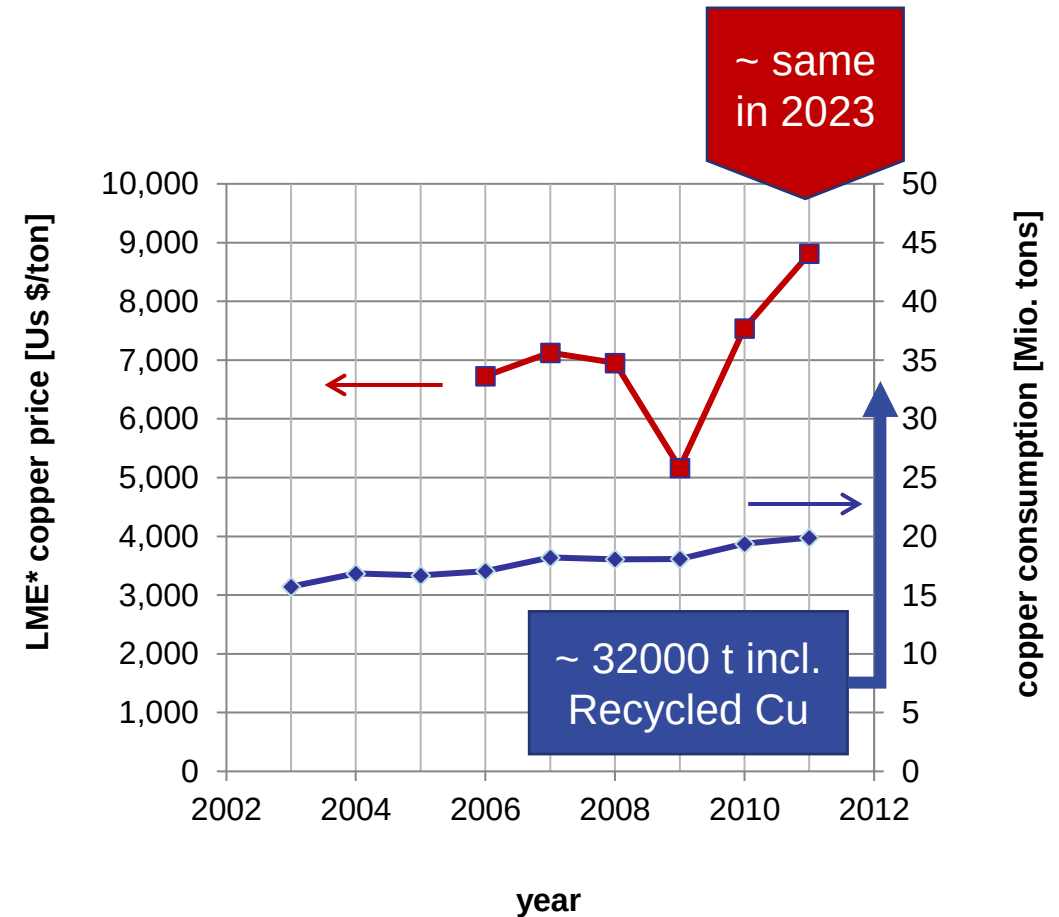
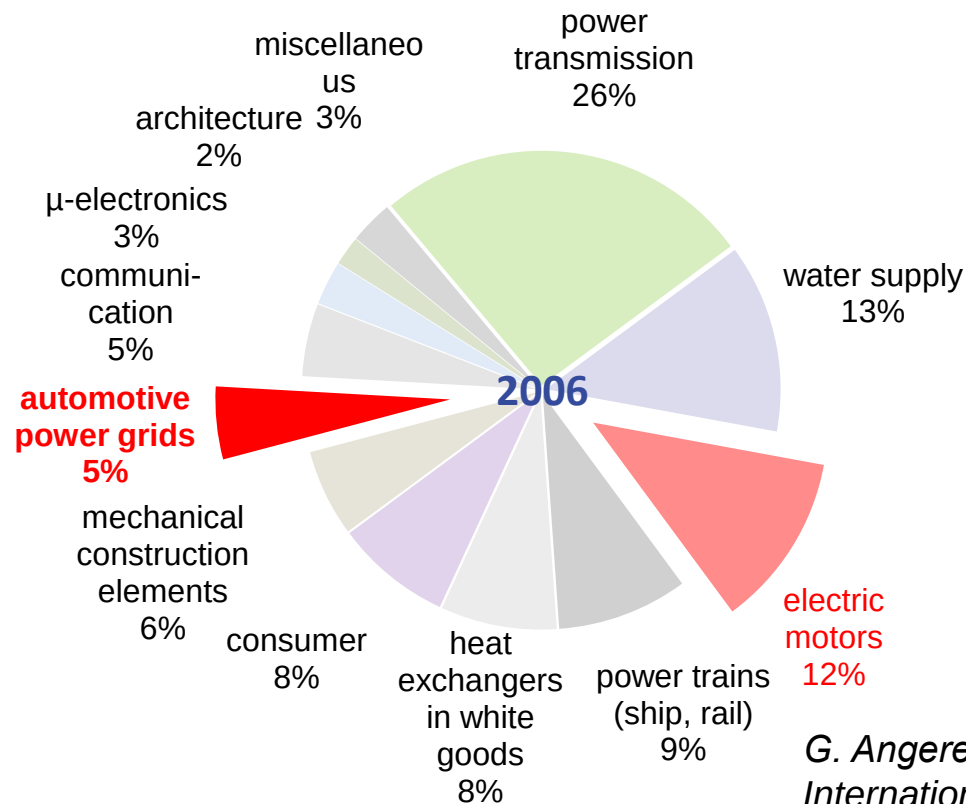
<https://www.eenewseurope.com/en/number-of-automotive-ecus-continues-to-rise/>

- 11.136 electrical parts in total
- communication:*
 - **61 ECUs in total**
 - external diagnosis for 31 ECUs via serial communication
 - optical bus for high bandwidth Infotainment-data
 - **sub-networks** based on proprietary serial bus
 - **35 ECUs** connected by **3 CAN-busses**
- sharing*
 - appr. **2500 signals**
 - in **250 CAN messages**

pictures: wikipedia.org

On-board power grid Cars

Copper markets worldwide



*LME: London Metal Exchange

G. Angerer et al, „Kupfer für Zukunftstechnologien“, FhG-ISI, Karlsruhe, Germany, and International Copper Study Group, The World Copper Factbook 2022

Power Grids in Buildings

Once upon a time...

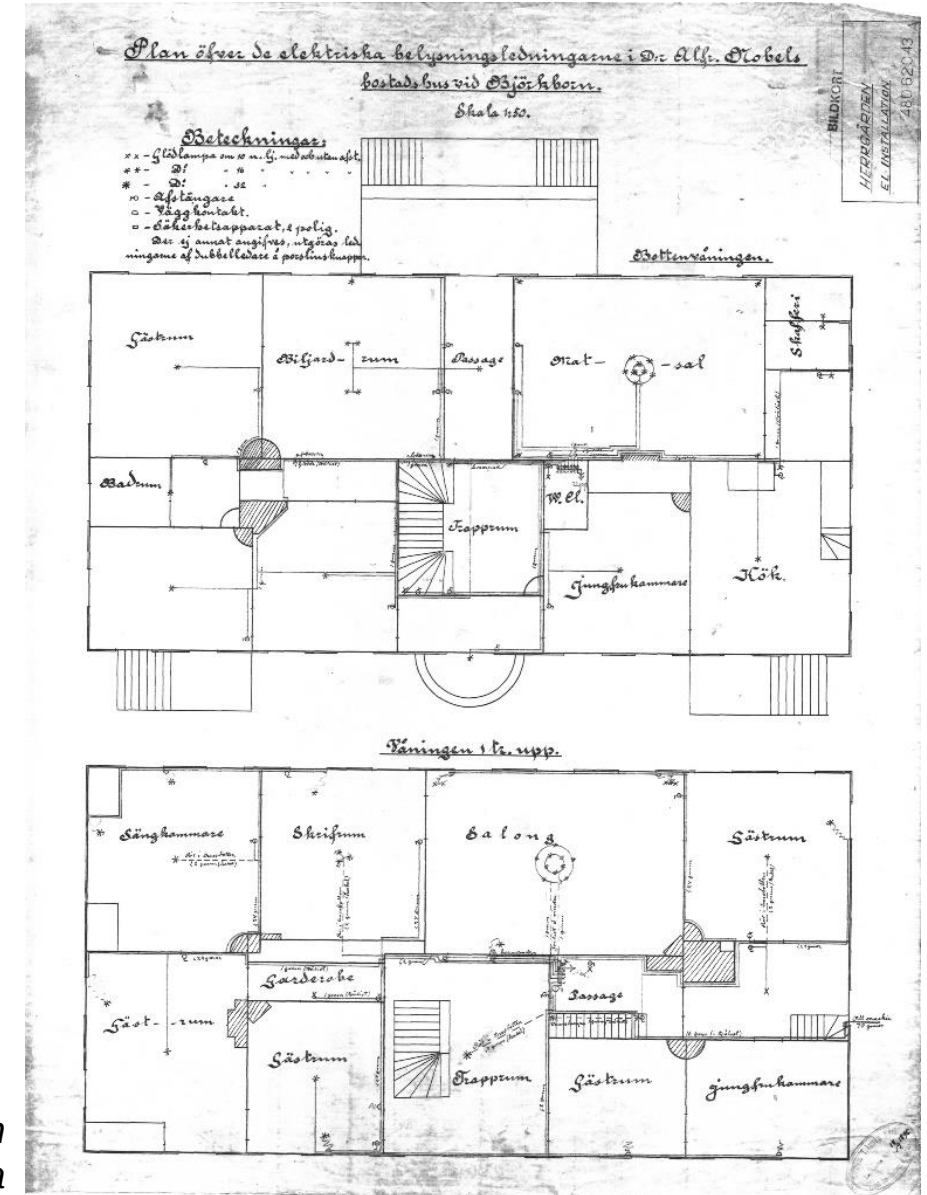
- ... buildings without automation!

19th century: **Power grid** as
„pure energy distribution network“

- simple circuitry (fuses, switches, light bulbs, wall sockets)
- low complexity
- relatively high degree of compatibility



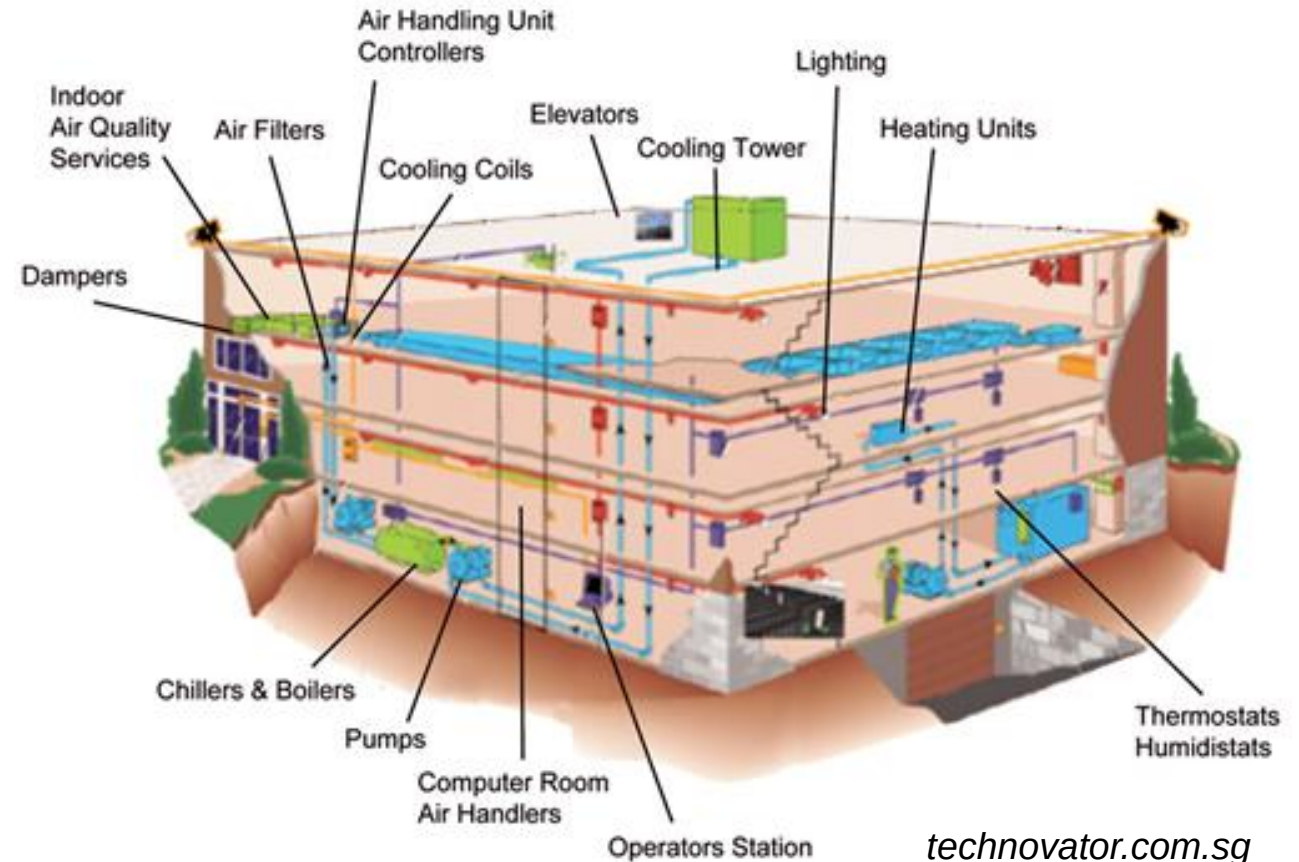
1889: Installation plan of Alfred Nobel's home in Karlskoga, Sweden © Nobel-Museum Karlskoga



Power Grids in Buildings

Building Automation

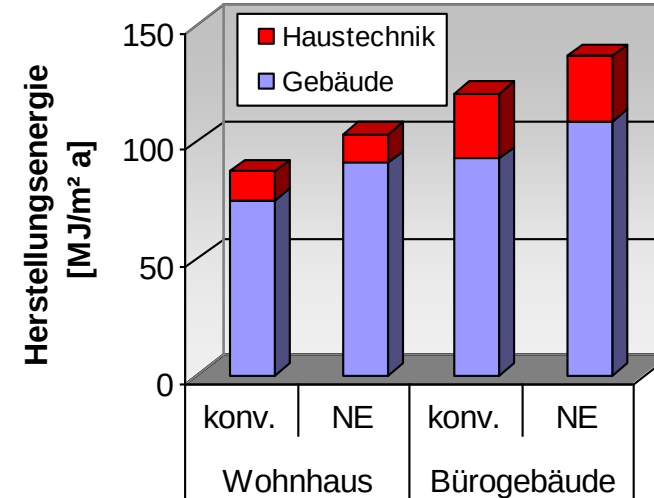
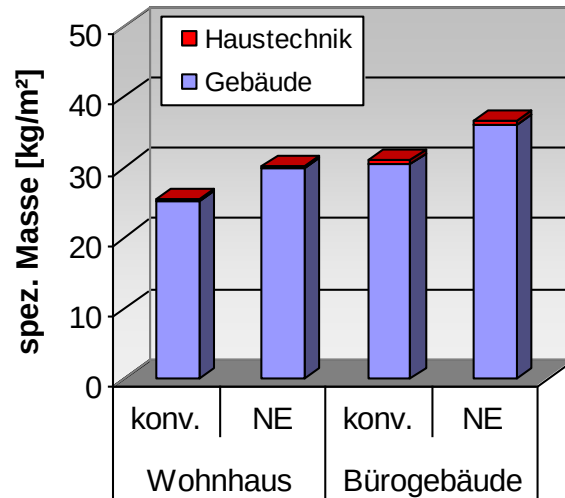
- Wishes and demands
 - alarm systems (fire, water, ...)
 - security and surveillance
 - room occupancy
 - building control (light, climate, ...)
 - comfort (smart home, ...)
- Technical realization
 - complex sensor and control systems
 - mostly wire or battery-based
 - severe incompatibilities (e.g. from different bus systems, ...)
 - manual installation and maintenance



technovator.com.sg

Power Grids in Buildings

Infrastructure for Building Automation



Amstein + Walther Ingenieure AG, Zürich, Umweltrelevanz der Haustechnik, Studie im Auftrag des Bundesamtes für Energie, Schweiz, 1998

Results

- The mass fraction of building appliances is small: ~ 1%
- The fabrication energy for building appliances is large: 15...30% (!)
- Additional costs due to maintenance and modifications are high

Power Grids in Buildings

„Grey energy“ of grid-based power supply

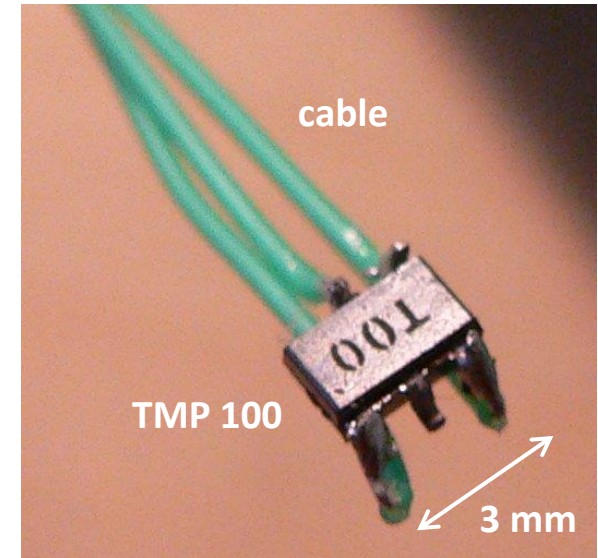
Example 1: A low-power temperature sensor module, supplied from a thin power cord with 1 m wire length

Questions

- How much “grey energy” is required to produce the connecting wires?
- How long could we supply the sensor from this energy budget?

Materials

- 3 x 1 m of 26 AWG (7/34) copper wire with PVC insulation (wall thickness: 0,010“), e.g. from Alpha Wire
- 1 digital temperature sensor, e.g. Texas Instruments TMP 100
power consumption: $100\ \mu\text{A}$ @ $3\ \text{V} = 300\ \mu\text{W}$



Power Grids in Buildings

„Grey energy“ of grid-based power supply

- **Embodied Energy or “grey energy” budget** (best case) of the connecting wires:

	specific fabrication energy [MWs/kg]	density [g/cm ³]	amount [kg]	fabrication energy [Ws]	fabrication energy [Wh]
Cu	55,7	8,96	0,00379	211.107,5	58,6
PVC	67,5	1,25	0,00131	88.166,4	24,5
				299.273,8	83,1

- Operational lifetime of the temperature sensor, supplied from this grey energy budget:

$$\frac{300.000 \text{ Ws}}{300 \text{ }\mu\text{W}} = 1 \cdot 10^9 \text{ s} = 31,7 \text{ years}$$

Power Grids in Buildings

„Grey energy“ of grid-based power supply

Example 2: Electric illumination is turned on and off with a conventional switch

Questions

- How much “grey energy” is required to produce the wires to/from the switch?
- What physical energy is applied to operate the switch?
- What electrical energy is required to transmit the information “light on/off”?



Image by Elgato

Materials and Energy

- 2 x 1 m of copper wire (cross section 0,75 mm²) with PVC insulation (wall thickness: 0.5 mm)
- mechanical energy required for pressing a switch: appr. 4.5 mWs
- electrical energy required to transmit the information “on/off”: appr. 150 μWs

Power Grids in Buildings

„Grey energy“ of grid-based power supply

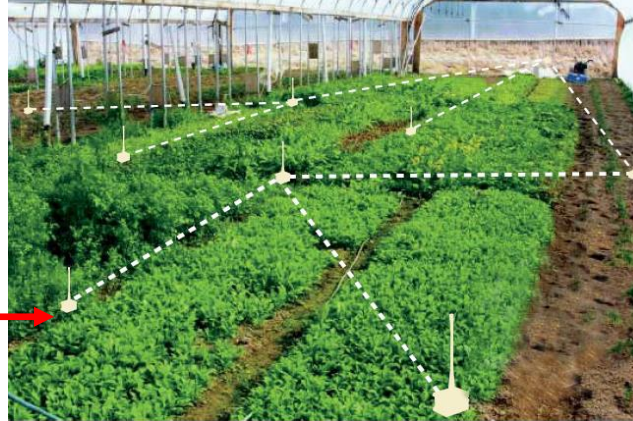
- Grey energy budget (best case) of the connecting wires:

	specific fabrication energy [MWs/kg]	density [g/cm ³]	amount [kg]	fabrication energy [Ws]	fabrication energy [Wh]
Cu	55,7	8,96	0,01344	748.608,0	207,9
PVC	67,5	1,25	0,00589	397.607,8	110,4
				1.146.215,8	318,4

- Comparison with the energies required for pressing a switch and for the wireless transmission of an „on/off“ signal

$$\begin{aligned} \text{switch operation:} & \quad \frac{1.146.216 \text{ Ws}}{4,5 \text{ mWs}} \approx 255.000.000 \text{ times} \\ \text{signal transmission:} & \quad \frac{1.146.216 \text{ Ws}}{150 \text{ }\mu\text{Ws}} \approx 7.641.000.000 \text{ times} \end{aligned}$$

What is the Best Option: Wire or Battery?

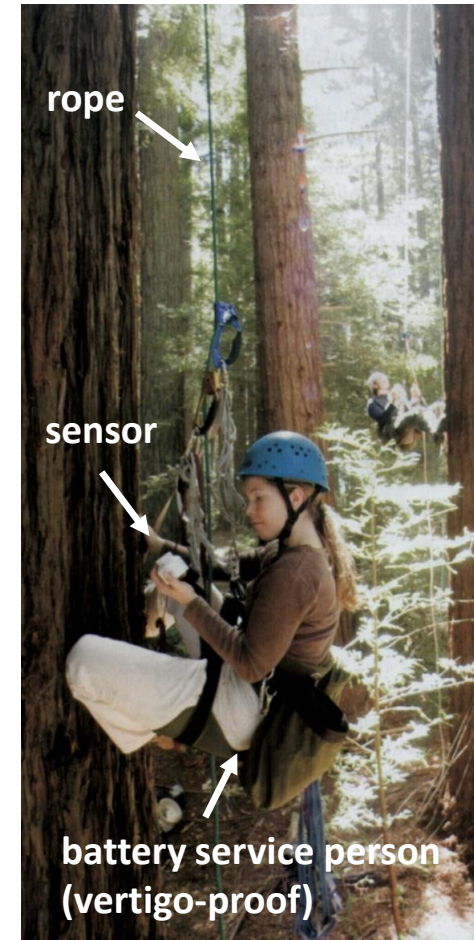


*distributed and „**embedded**“ sensor systems in greenhouses © Crossbow*



*Medical implants
© Vitatron*

tire pressure sensors



*Sensors in
redwood trees
© University of
California*

The Vision

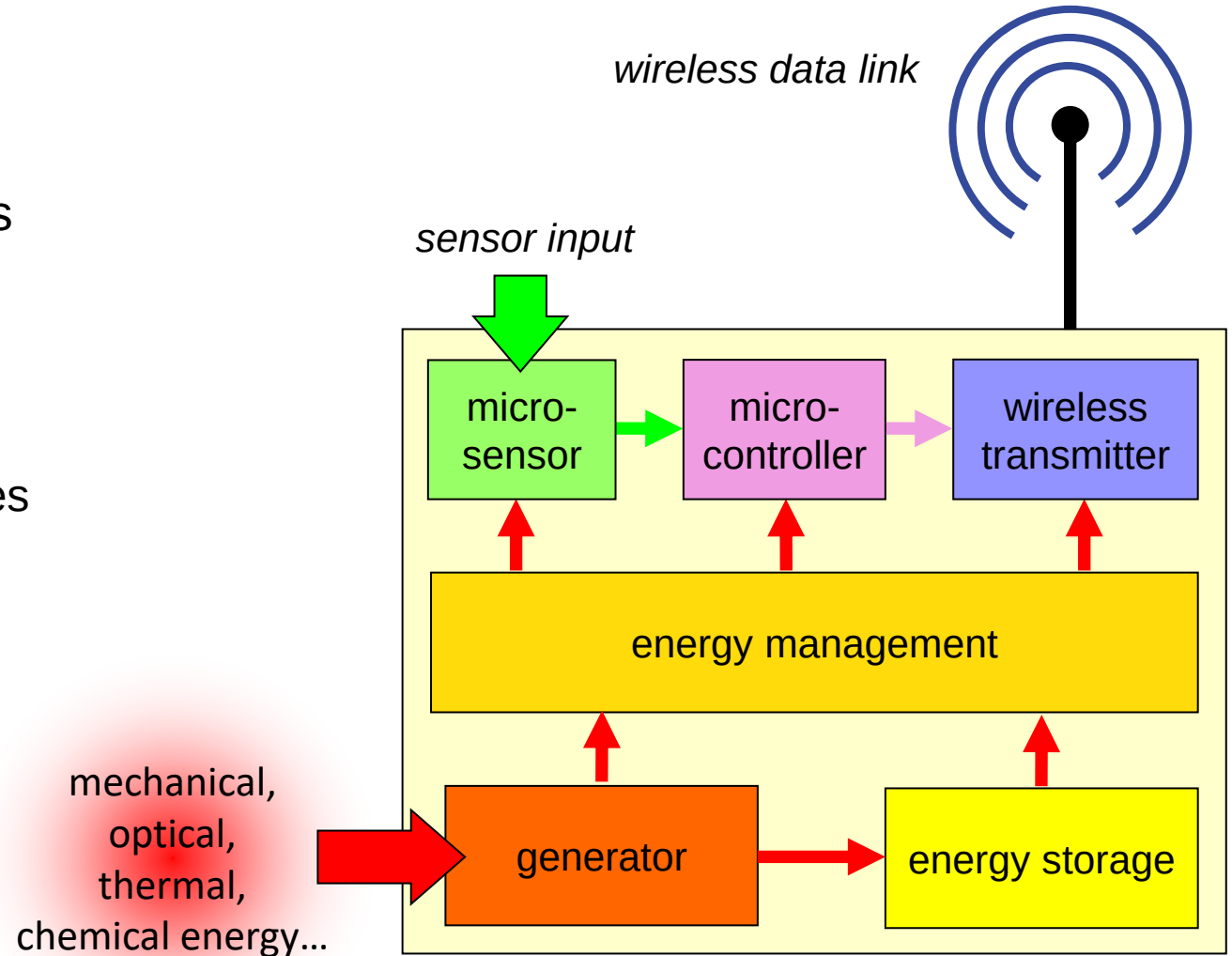
Micro Energy Harvesting

Energy-Autonomous Embedded Systems

- „always on“
- no battery recharging or exchange
- no power cords
- easy to install at numerous application sites

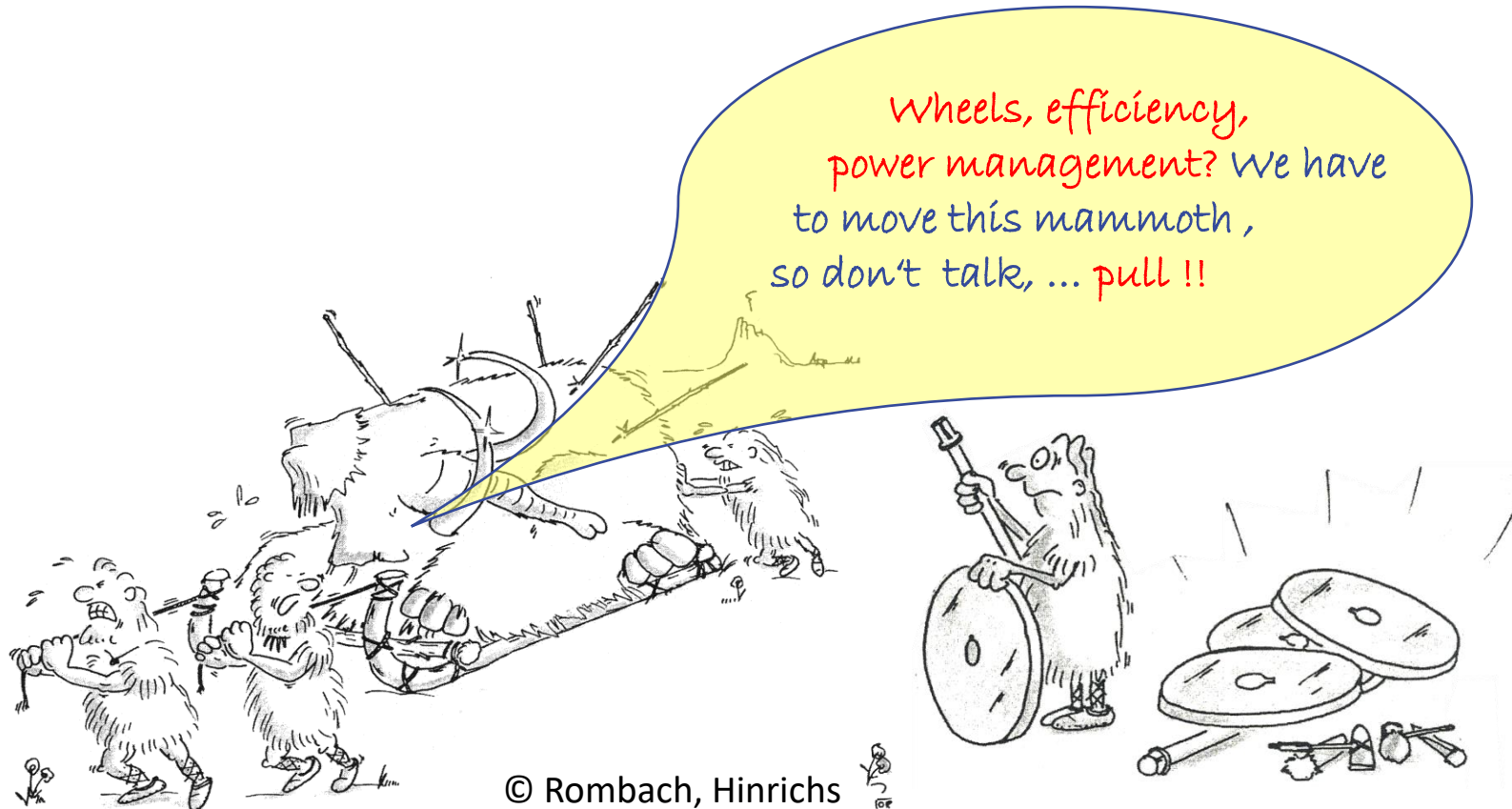
Micro Energy Harvesting

- Energy supply concept for embedded microsystems
- conversion of local „ambient“ energy into electrical energy
- Goal: energy-autonomous operation of embedded systems



Micro Energy Harvesting

How?



Micro Energy Harvesting

Misconceptions

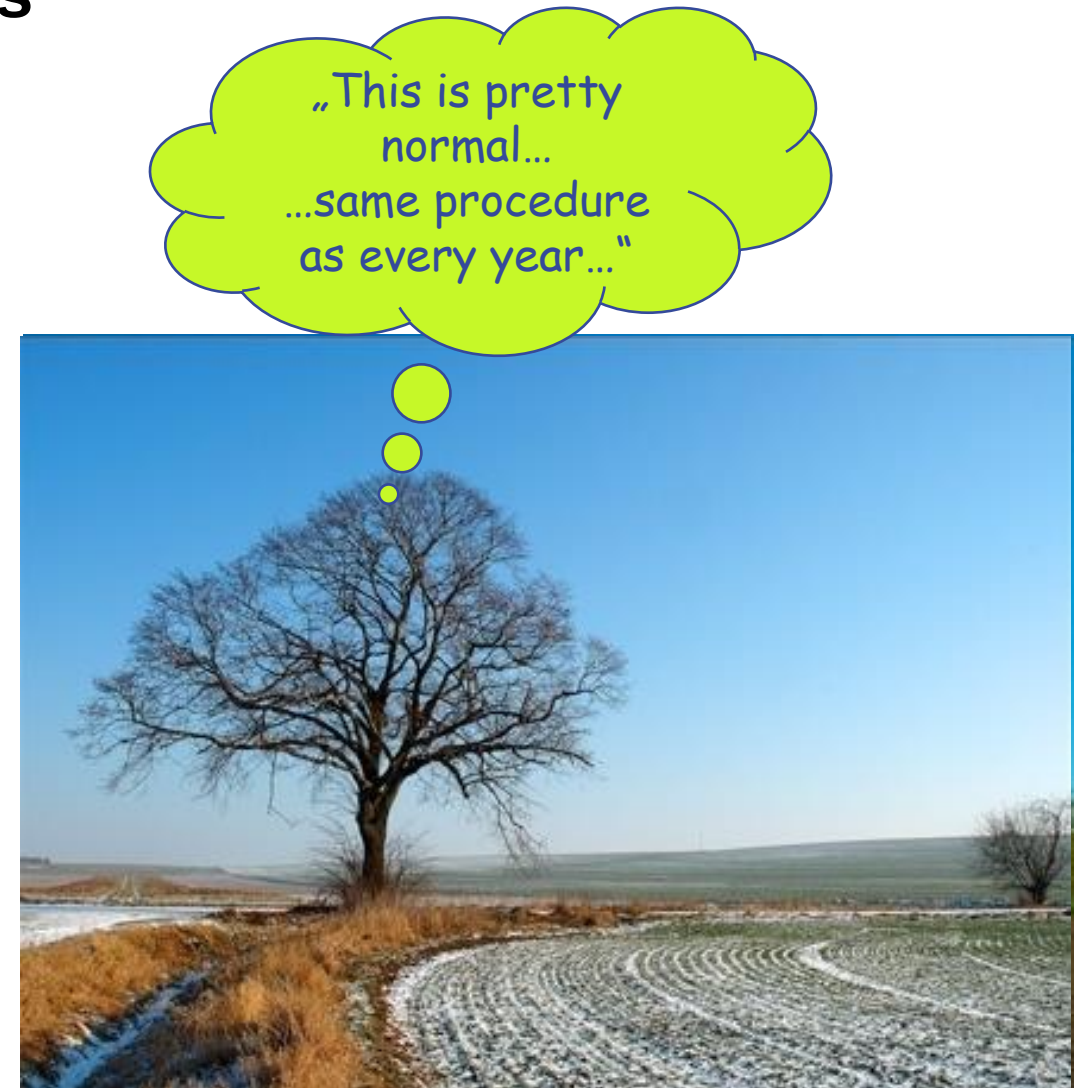
- (some) **Journalists:** „Micro Energy Harvesting will stop the climate change.“
- **Politician:** „Micro Energy harvesting is - on a small scale - the same as we do with regenerative energies - on a large scale.“
- **Consumer:** „Energy harvesting is nothing else than replacing the battery with a generator.“
- **System designer:** „I have a sensor that requires 24 V and 50 mA.
If you can supply that, we will use energy harvesting.“
„The 100 μ W that you deliver with your harvester will not be sufficient for our system. Forget about it.“

Micro Energy Harvesting

General Requirements and Constraints

We have to operate an embedded system under ...

- highly variable ambient power
day-night, summer-winter, indoor-outdoor, on-off times of technical environments, variable temperatures, vibration amplitudes and frequencies, irradiation levels...
- low-energy and zero-energy phases
„there is no energy to harvest from“
- variable internal power consumption
sensor and data processing: (very) low
wireless communication: very high
- undesirable environmental influences
temperature, humidity, vibration level,...



Micro Energy Harvesting

What do we REALLY want to accomplish?

Requirements from the product/application designer:

- a certain system function,
- with a certain level of reliability,
- under other requirements, e.g. costs, lifetime, material choice,...
- at a certain application site.

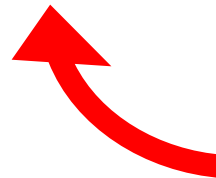
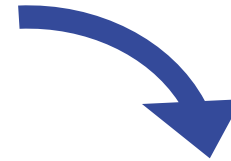
Requirements from the product/application designer:

First: retrieve knowledge about the application site

- available ambient energy
- allowed space, mass ...

Then: design of ...

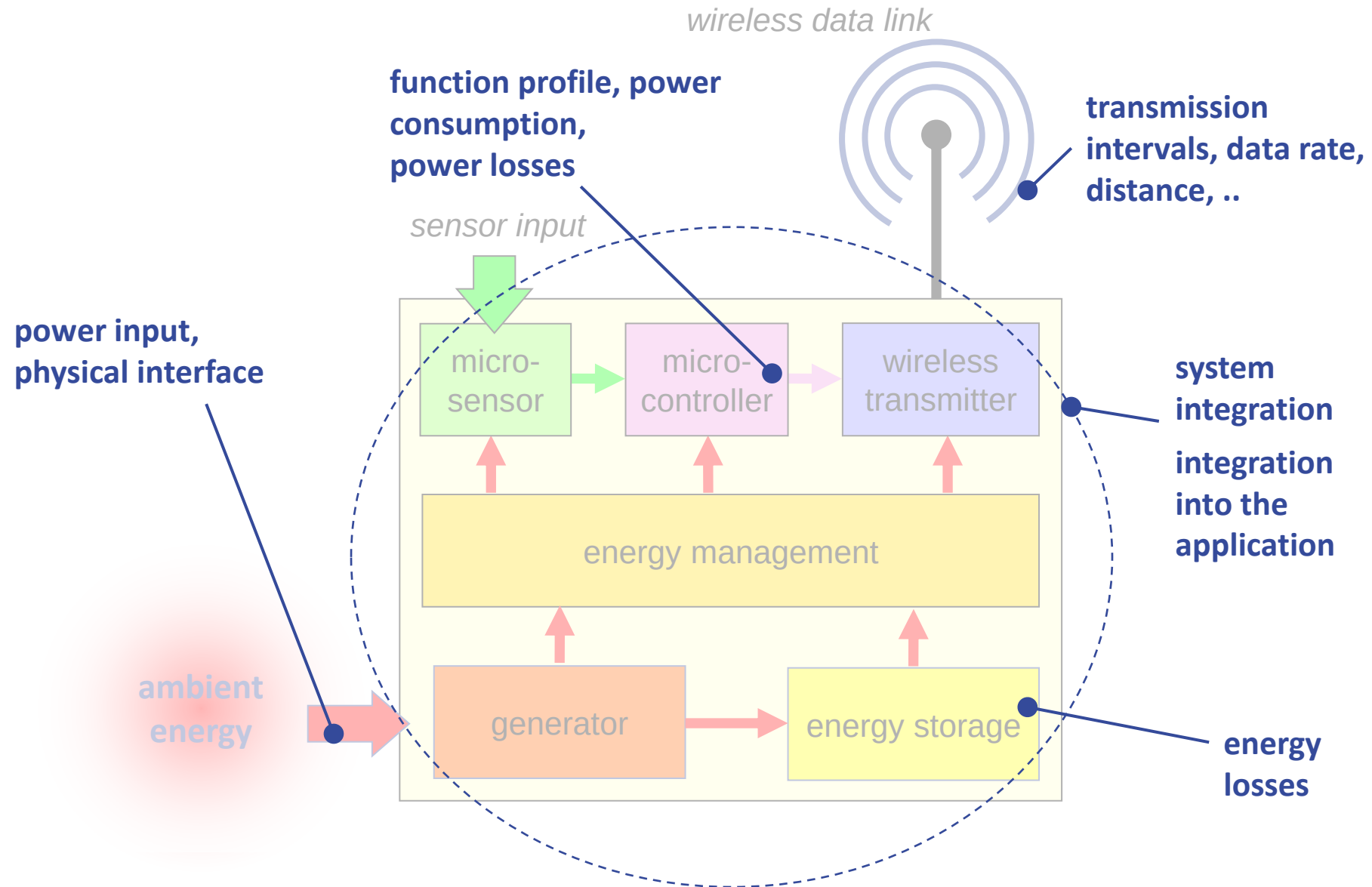
- Generator
- energy storage
- power management
- electronic system
- data transmission
- system concept
- system integration



A complex system design!

What to do? ...

System Design from a Global Perspective



Micro Energy Harvesting

Global Power Calculation

Power Input

1. type of ambient energy?
2. type of generator?
3. type of physical interface?

4. Input power

- minimum (usable) input power?
- maximum input power?
- power profile over time?
- required input power?

➡ **effective input power (EIP)**

$$EIP = S \cdot EPC$$

S : safety margin (> 1)

Power Consumption

1. required functions and reliability?
2. required application site?

3. Power consumption

- sensing or actuation
- data processing
- communication
- power management
- system functions (e.g. timers)
- start-up/shut-down, sleep, wait, ...

functional

auxiliary

➡ **effective power consumption (EPC)**

Power Balance

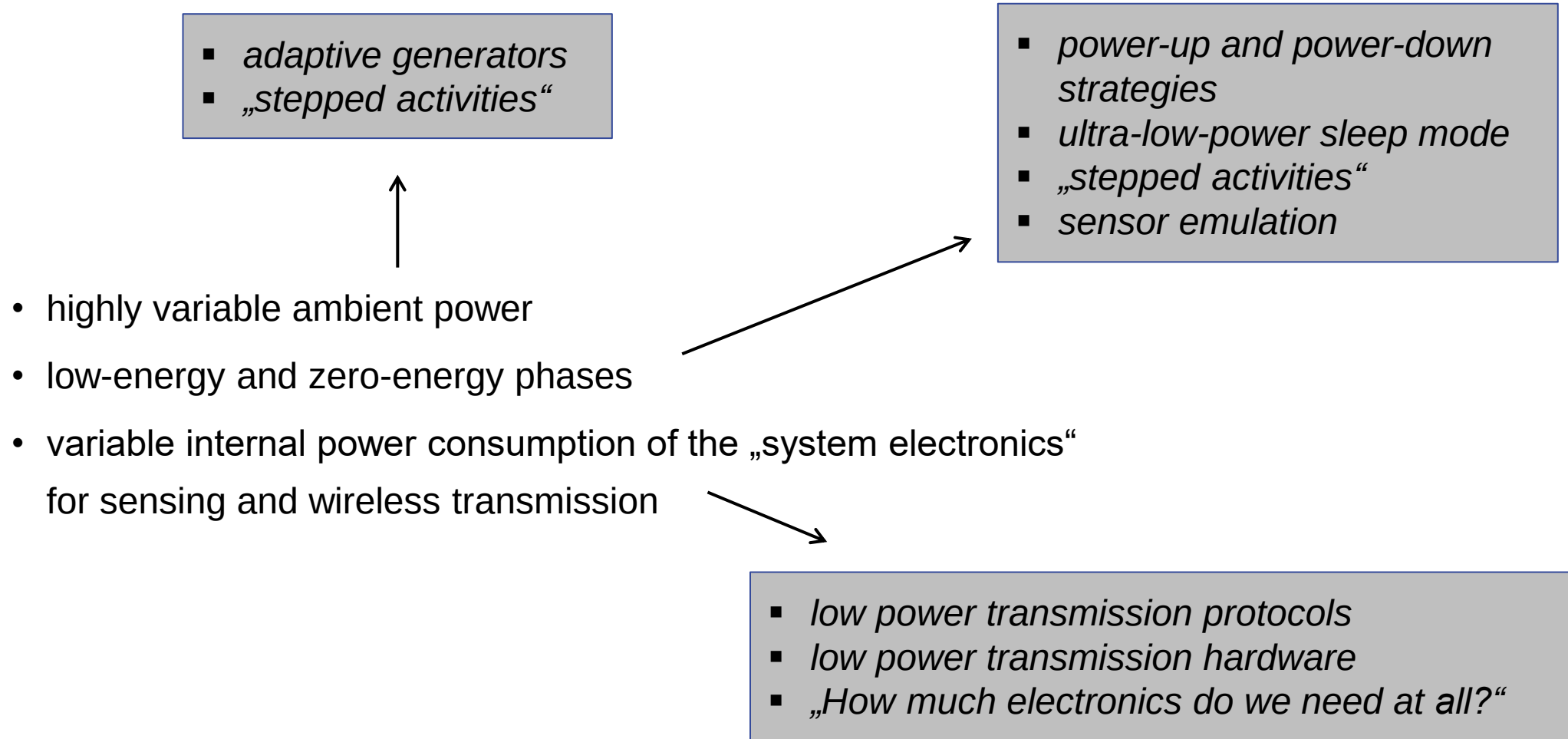
Power generation (EIP) →	Power consumption (EPC) ↓	[mW]	Shoe generator (Kymissis)	Electromotor used as generator	Hand cranking radio (Sony)	Bycicle dynamo (AXA HR)	Stepcharger foot step generator	Cycling bike generator
			230	1.000	1.650	3.000	6.000	150.000
Swatch watch	0,001		230.000	1.000.000	1.650.000	3.000.000	6.000.000	150.000.000
Travel clock	0,002		115.000	500.000	825.000	1.500.000	3.000.000	75.000.000
Hearing aid	0,05		4.600	20.000	33.000	60.000	120.000	3.000.000
Cardiac pacemaker	0,10		2.300	10.000	16.500	30.000	60.000	1.500.000
Tamagotchi	0,36		639	2.778	4.583	8.333	16.667	416.667
Philips pager Fiori (1998)	2		115	500	825	1.500	3.000	75.000
LED pointer	3		77	333	550	1.000	2.000	50.000
Volvo S80 car remote control	12		19	83	138	250	500	12.500
Nokia 6185 cell phone (1999)	18		13	56	92	167	333	8.333
Motorola Startec 130 cell phone(1999)	37		6	27	45	81	162	4.054
Philips Genie cell phone (1997)	48		5	21	34	63	125	3.125
Philips portable radio AE6545	52		4	19	32	58	115	2.885
Sony radio TCF-B200	60		4	17	28	50	100	2.500
Philips Fizz cell phone (1996)	66		3	15	25	45	91	2.273
LED flashlight (torch)	80		3	13	21	38	75	1.875
Freeplay FPR1 radio	90		3	11	18	33	67	1.667
Philips cordless FM headphone	120		2	8	14	25	50	1.250
Philips MP3 player (2000)	220		1	5	8	14	27	682
Philips cassette player ACT6688	276		0,83	4	6	11	22	543
Philips CD player ACT7582	432		0,53	2	4	7	14	347
Philips MP3 player (2000)	440		0,52	2	4	7	14	341
Motorola Iridium satellite phone	570		0,40	2	3	5	11	263
Philips Velo 1 PDA	600		0,38	2	3	5	10	250
Incandescent flashlight	700		0,33	1	2	4	9	214
Motorola 130 (during transmission)	720		0,32	1	2	4	8	208
Halogen flashlight (torch)	2.000		0,12	0,50	0,83	2	3	75
Philips Xalio mobile phone (1997)	2.703		0,09	0,37	0,61	1	2	55
Philips mini-notebook (Windows CE)	3.400		0,07	0,29	0,49	0,88	2	44
Video 8 camera	6.000		0,04	0,17	0,28	0,50	1	25
Notebook computer	15.000		0,02	0,07	0,11	0,20	0,40	10
DVD player	20.000		0,01	0,05	0,08	0,15	0,30	8
Personal computer	35.000		0,01	0,03	0,05	0,09	0,17	4
TV (30 cm)	50.000		0,00	0,02	0,03	0,06	0,12	3

$$S = \frac{EIP}{EPC}$$

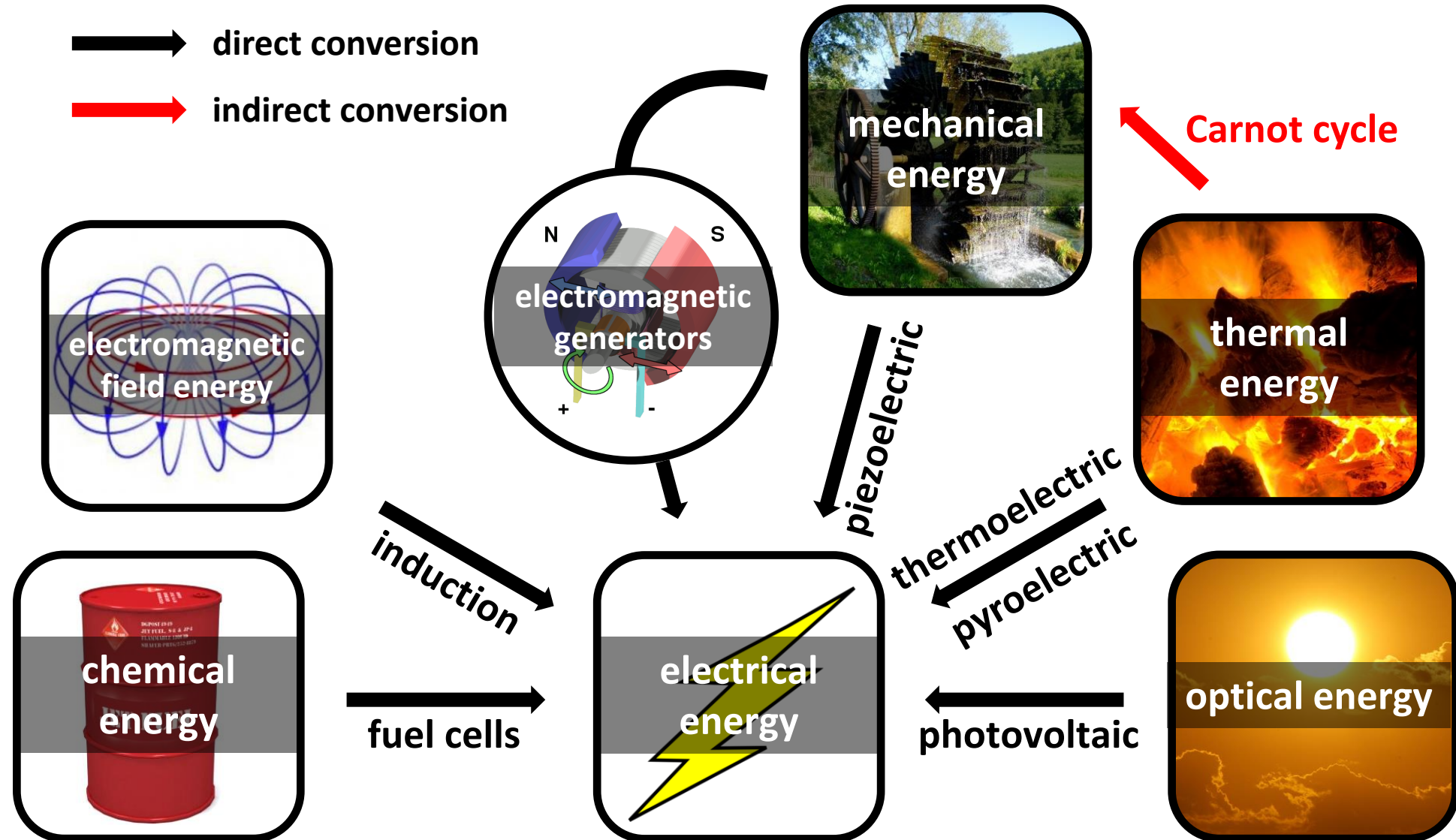
De Pater, TU Delft (2000)

Micro Energy Harvesting

Appropriate System Concepts

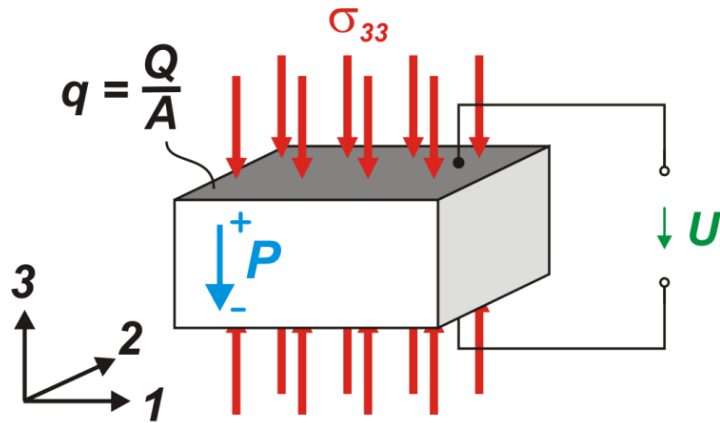
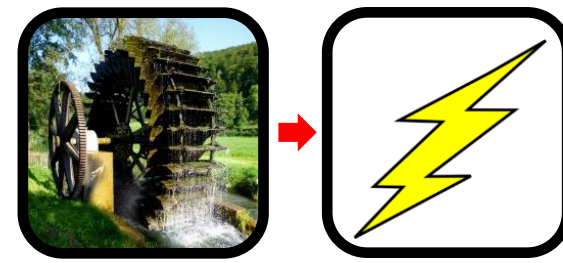


Ambient Energies and Conversion Mechanisms



Ambient Energies and Conversion Mechanisms

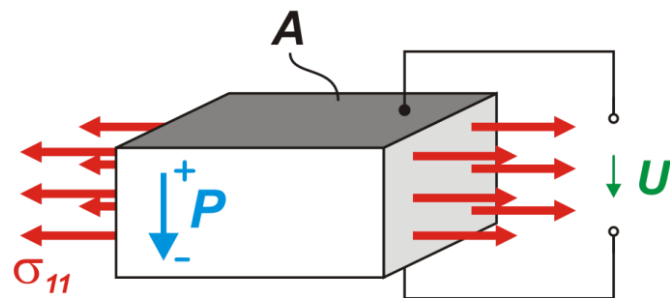
Piezoelectric Generators



vertical mode: $Q = d_{33} \cdot \sigma_{33} \cdot A$

Piezoelectric constants of relevant materials

	d31	d33 [pC/N]
PZT	- 350	680
LiNbO ₃	-0,85	6
PVDF	6..10	13..22



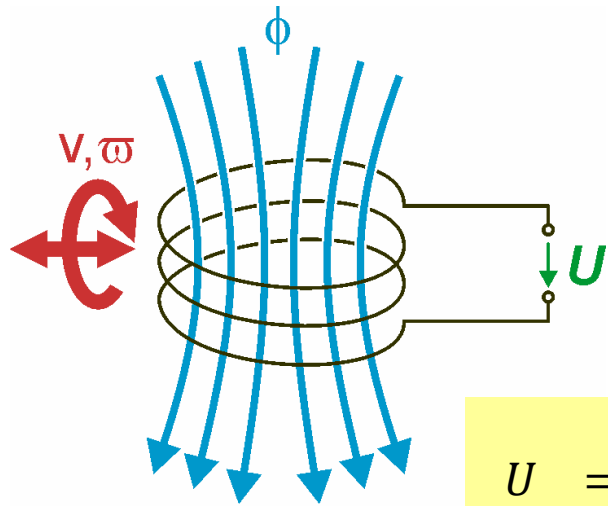
transversal mode: $Q = d_{31} \cdot \sigma_{11} \cdot A$

Properties

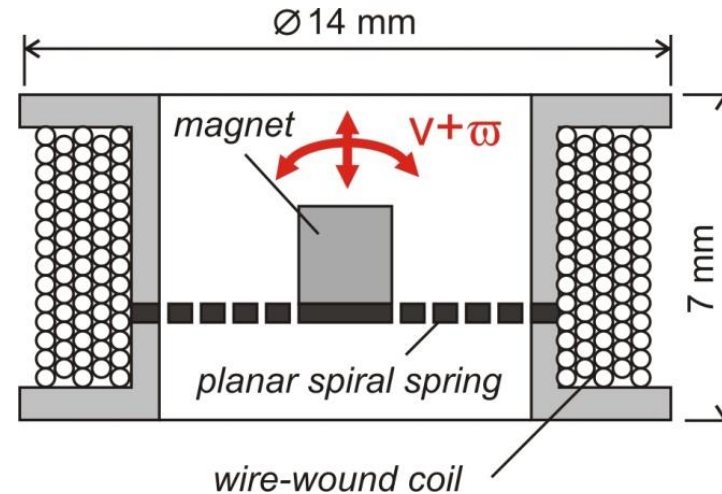
- charge delivery
- AC currents only via a dynamic mechanical loading
- fair to high voltages (1...100 V)
- moderate to high source impedance (10 ...100 kΩ)

Ambient Energies and Conversion Mechanisms

Electromagnetic Generators

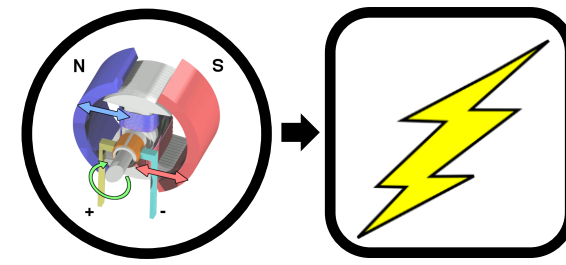


$$U = -N \cdot \frac{d\Phi}{dt}$$



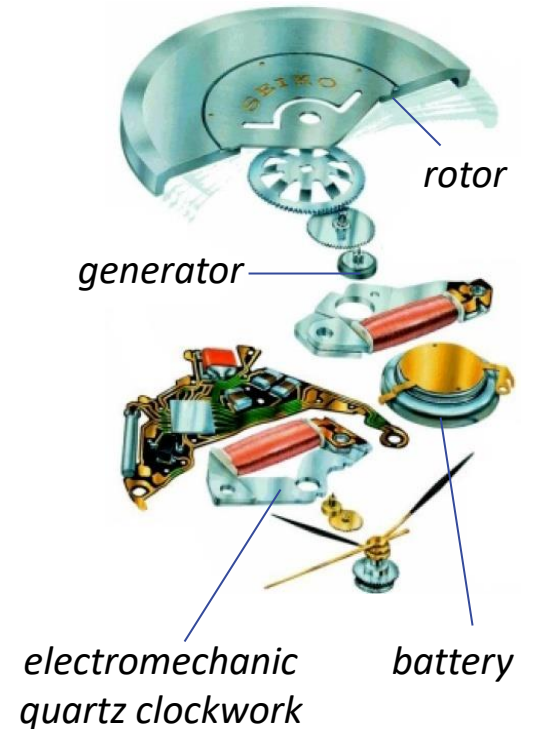
multi-resonant generator

Univ. Hongkong, 2002



Properties

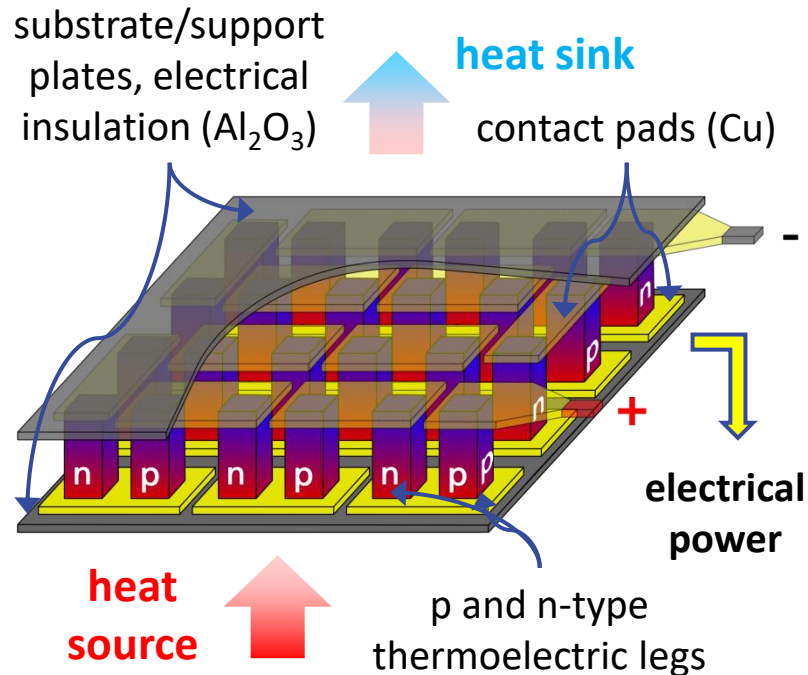
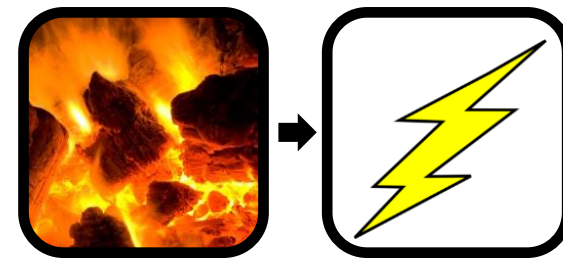
- AC currents from motion or induced AC fields
- bad to fair voltage range (mV...V)
- moderate source impedance ($<10 \text{ k}\Omega$)



rotatory generator of the Seiko Kinetic™ wrist watch

Ambient Energies and Conversion Mechanisms

Thermoelectric Generators (TEGs)



$$\Delta U = \alpha \cdot \Delta T$$

Seebeck coefficients of some material combinations

	α [$\mu\text{V/K}$]
Al / p-Poly-Si	195
Al / n-Poly-Si	110
p-Poly-Si / n-Poly-Si	190...320
p-Bi _{0,5} Sb _{1,5} Te ₃ / n-Bi ₂ Te _{2,85} Se _{0,15}	200...420

Properties

- DC-like currents, however...
- polarity changes with the direction of ΔT
- low to fair voltages (100 mV ... V)
- low to high load resistance (Ω ... $\text{M}\Omega$)

Ambient Energies and Conversion Mechanisms

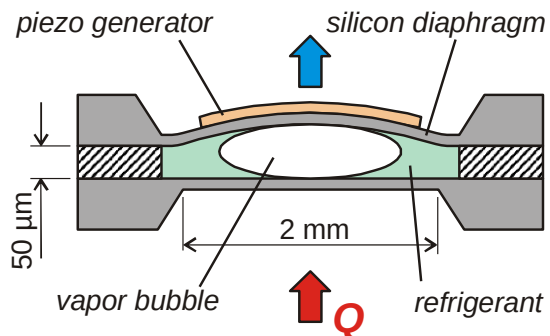
Micro Heat Engines



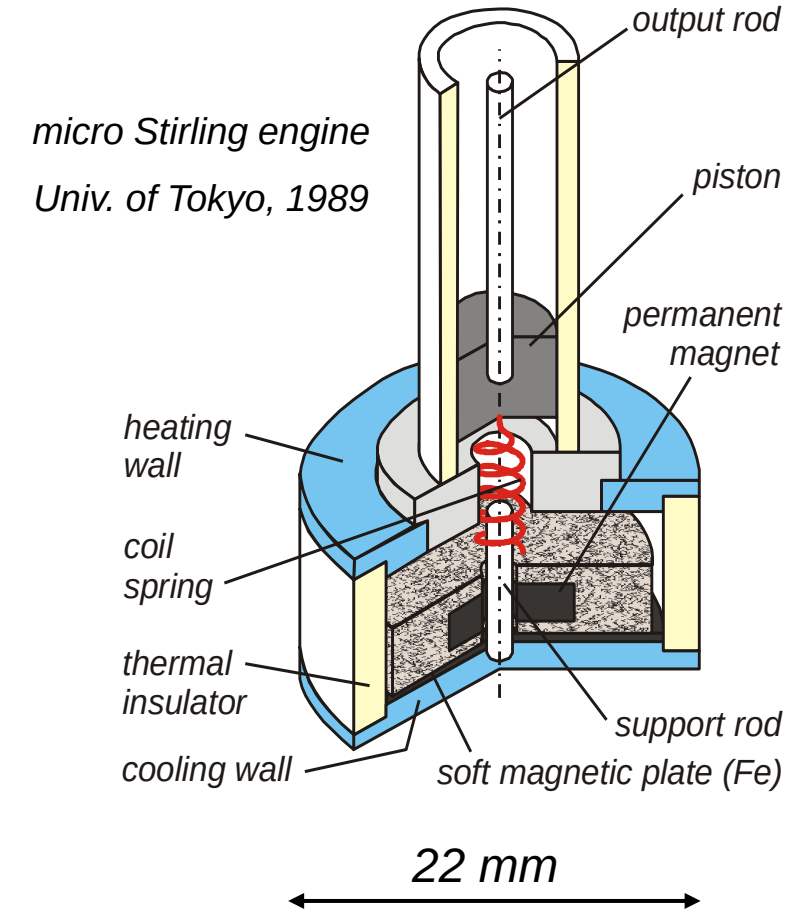
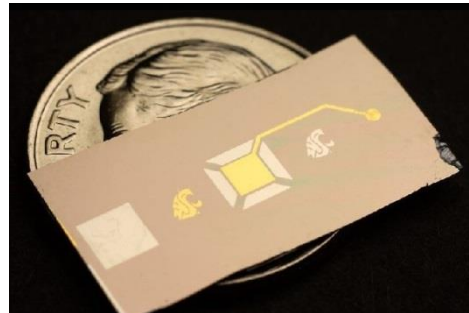
Properties

- use of thermodynamic cycles
- potentially high efficiency (Carnot), but ...
- potentially high losses
- requires a mechatronic conversion

$$\eta_c = \frac{W}{Q} = 1 - \frac{T_0[K]}{T_1[K]}$$

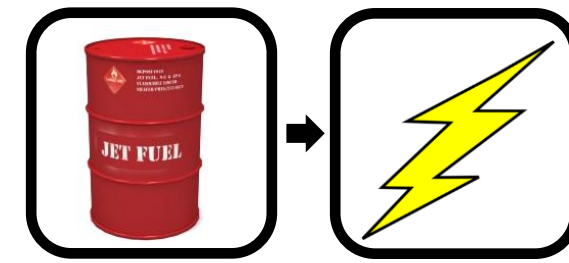


P³ two-phase heat engine, WSU, Seattle, USA, 2003



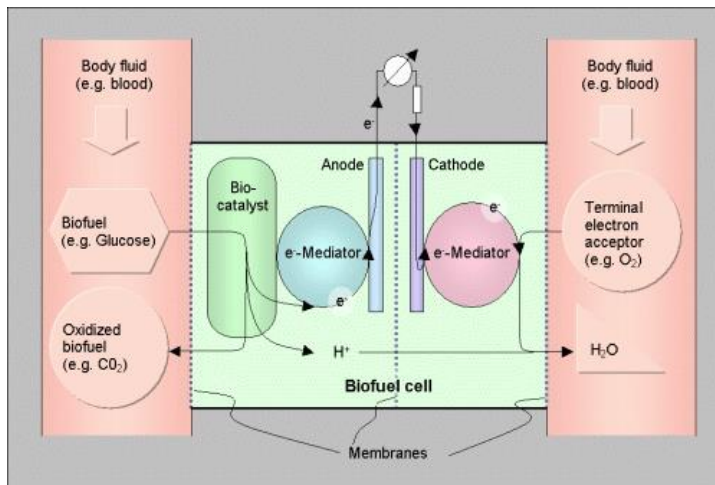
Ambient Energies and Conversion Mechanisms

Bio Fuel Cells



Properties

- DC current from catalytic oxidation of a „bio fuel“
- use of different (bio)catalyzers enzymes, bacteria, metals
- low voltages (0.1 ... 0.5 V)
- power in the μW ... mW range

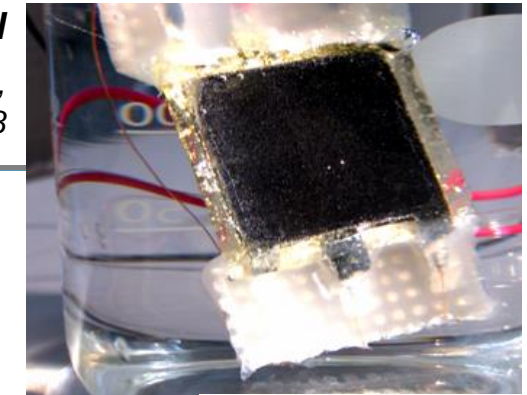
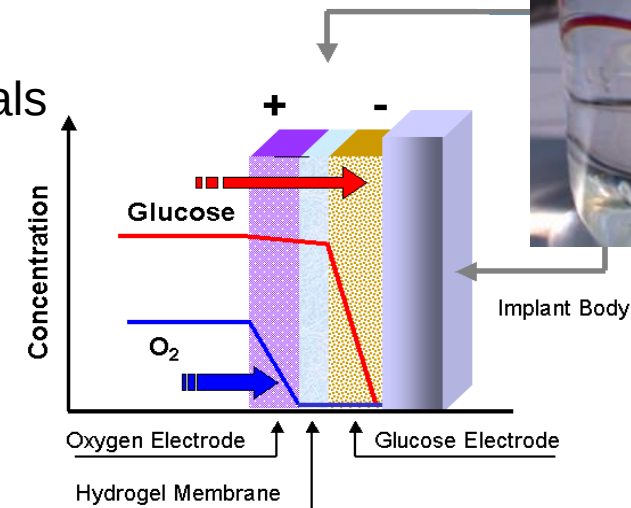


**enzymatic
bio fuel cell**

Sony, 2007

catalytic glucose fuel cell

S. Kerzenmacher et al.,
J. Pow. Sources, 2008



PREDATORY ROBOTICS

Microbial fuel cells digest flies to generate electricity for EcoBot II

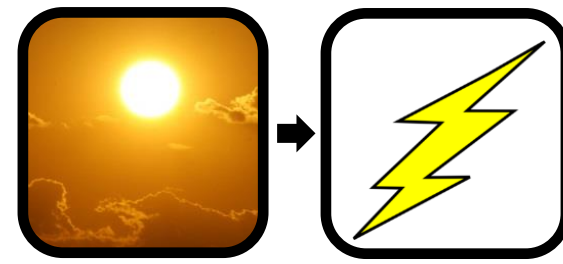


Microbial bio fuel cell

ECOBOT II, 2004

Ambient Energies and Conversion Mechanisms

Photovoltaic Cells

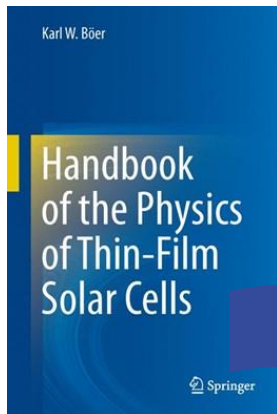


Thin film cells

- deliver DC voltage
- thickness: 0.3...0.6 mm
- flexible and washable
- typical efficiency : 3 %

sun light: $\approx 3.0 \text{ mW/cm}^2$

indoor light: $\approx 0.1 \text{ mW/cm}^2$



430 €



Si-thin film cell on a polymer carrier © altenergy.org



© Dethleffs

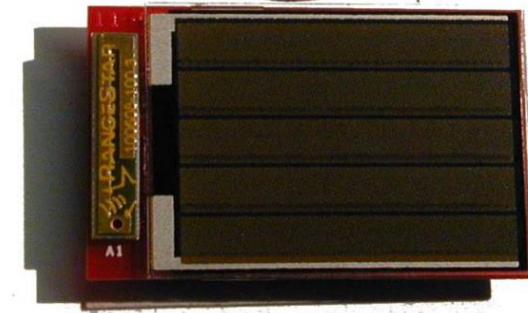
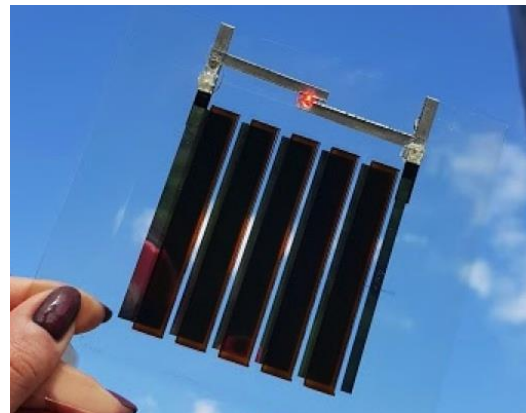


Si/perovskite thin film cell on glass carrier

© altenergy.org

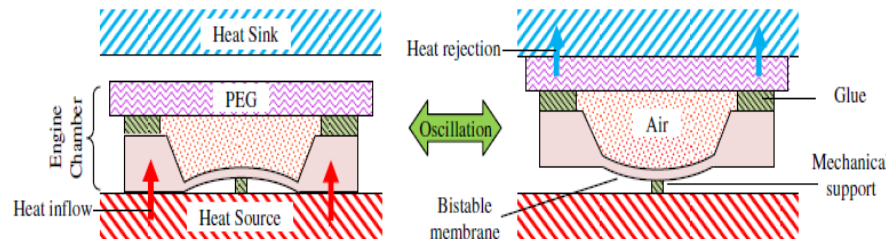
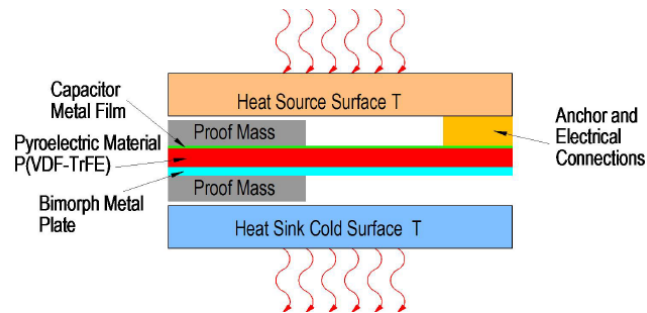
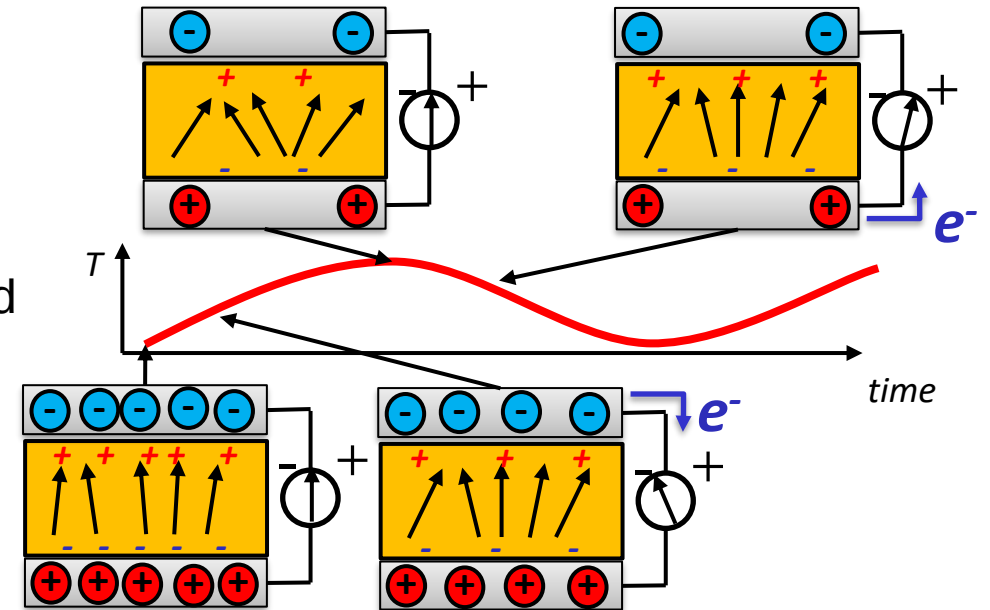
Si-thin film cell on a glass carrier

© Panasonic



A diagram illustrating the process of simplification. On the left is a square icon with rounded corners and a black border, containing a detailed, high-resolution image of a volcanic eruption with bright orange and yellow lava flows and dark, jagged rock formations. A black arrow points from this icon to a second icon on the right. The second icon is also a square with rounded corners and a black border, but it is white and contains a simple, stylized yellow lightning bolt with a black outline, representing a simplified or abstracted version of the original image.

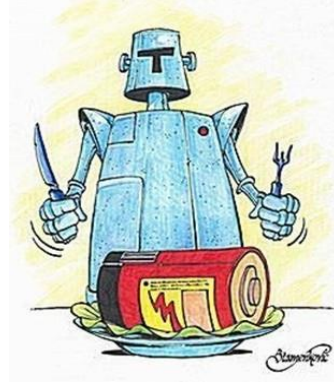
$$I_p = p_c A \frac{dT}{dt}$$



S. K. T. Ravindran et al., J. Micromech, 2012

System Components

Energy Storage

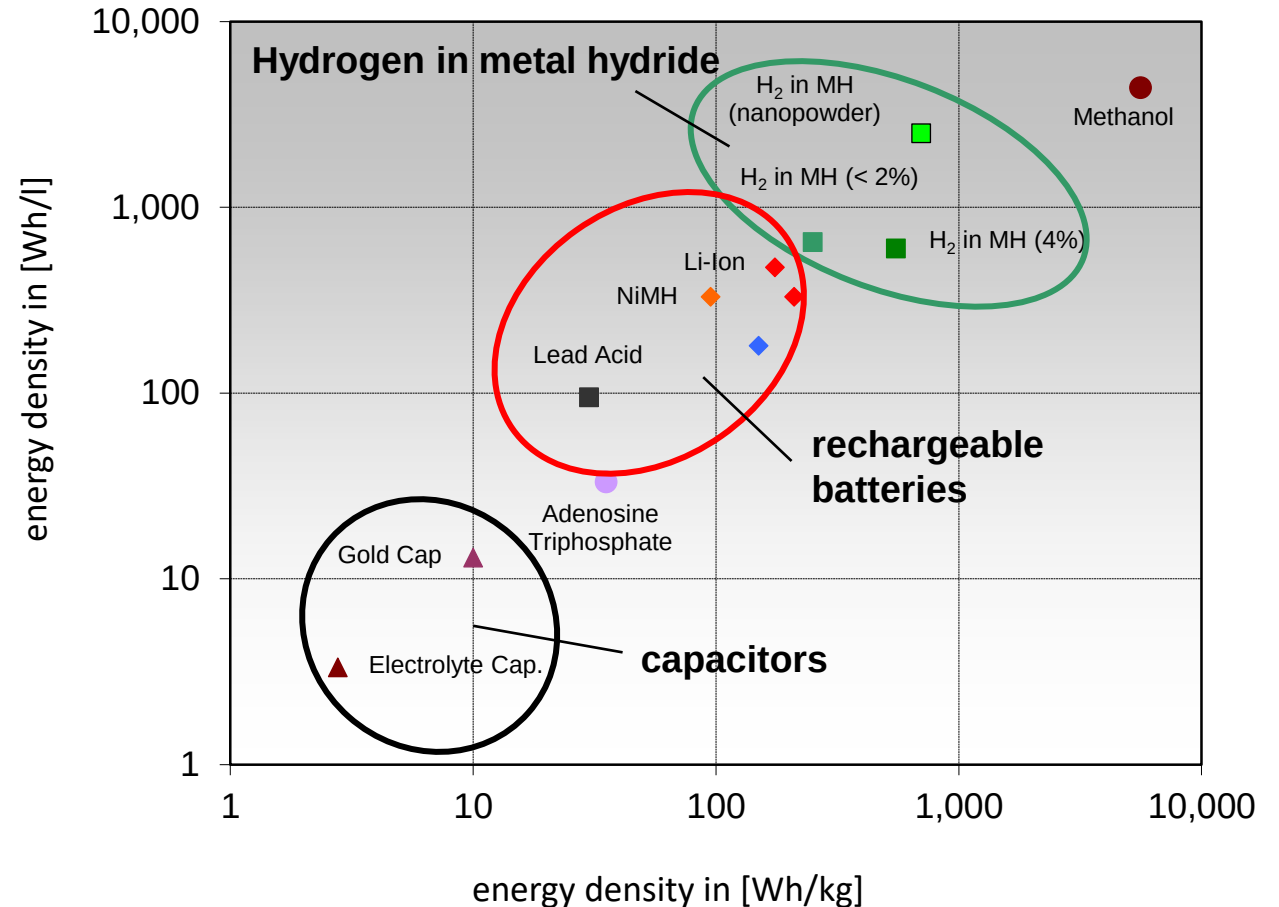


Global properties

- today mostly chemical (batteries)
- slowly emerging alternatives
- design conflict between storage capacity, size/weight and lifetime (always)

Requirements

- good Faraday efficiency
- appropriate size, weight and capacity
- appropriate cycle number (charge/discharge)
- small self-discharge and ageing effects
- others (e.g. temperature tolerance)



Refs: J. Brodd et al, J. Electrochem. Soc., 151 (3), 2004, K1-K11 and HERA Hydrogen Storage Solutions, D

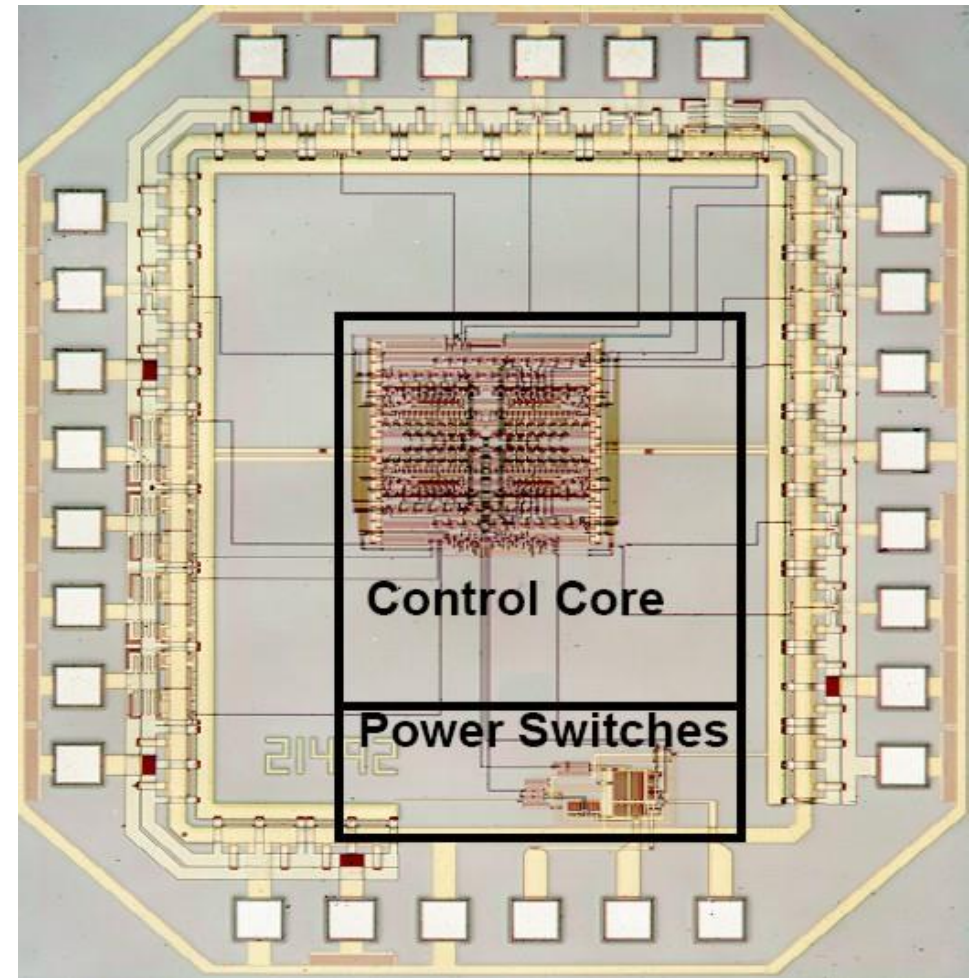
System Components

Energy Management

Requirements

- optimal impedance match between generator, battery and load
- voltage level transformation
- active generator control
- active rectification
- defined startup/shutdown
- supply voltage: $< 1\text{ V}$
- power consumption: few μW

Control ASIC for a capacitive micro converter,
Medinger, Ph. D. thesis, MIT &
Analog Devices, 1999



← 2163 μm →

Conclusion

- Micro Energy Harvesting as power supply for distributed, embeded systems...
- ... as additional option surpassing / supplementing battery and grid-based power supplies, if...
- ... we can design a system capable of
 - harvesting highly variable ambient power
 - living through low-energy and zero-energy phases
 - handling variable internal power consumption during sensing and transmission phases

→ Be a bit like a tree!

